

ABBREVIATIONS AND ACRONYMS

Lao PDR	Lao People's Democratic Republic
UNDP	UNITED NATIONS DEVELOPMENT PROGRAMME
MOF	The Ministry of Financial
MOIC	Ministry of Industry and Commerce
MPI	Ministry of Planing and Investment
APIS	Consultant APIS Co. Ltd
TaxETS	Tax Exemption Tracking System for the Ministry of Finance
GUI	Graphical User Interface
API	Application Interface
FPD-MOF	Fiscal Policy Department, Ministry of Finance
MPI	The Ministry of Planning and Investment
MOLSW	Ministry of Labor and Social Welfare
SEZ	Special Economic Zone
SEZO	Special Economic Zone Office
ASYCUDA	Automated Customs Data System
TaxRIS	Tax Revenue Information System
TE	Tax Expenditures
PT	Profit Tax TE
VAT	Value added tax must be paid

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1 Introduction

This report presents the progress and outcomes achieved under Deliverable 4 of the Tax Expenditures Tracking System (TaxETS) project, as stipulated in the Advisory Services Agreement between the Ministry of Finance (MoF) and APIS Company Limited. Deliverable 4 represents a major milestone: the completion of core development, interface refinement, and comprehensive testing of the TaxETS platform.

Building on the Inception Report (Deliverable 1), Blueprint Report (Deliverable 2), and Database Testing and Improvement (Deliverable 3), this report focuses on system enhancements implemented following feedback from the Fiscal Policy Department (FPD), the Ministry of Finance Expert(s), and UNDP. The improved system is now accessible to FPD and UNDP colleagues. Access credentials have been shared separately.

The report highlights key achievements in testing and refinement, explains the strategic adjustment to the implementation schedule, and provides evidence that the system now accurately calculates tax expenditures across multiple tax types. .

2 Background and Objective

The Tax Expenditures Tracking System (TaxETS) is developed under the National Planning and Financing for Inclusive Development (NPFID) project to strengthen the Ministry of Finance’s capacity to monitor, record, and evaluate tax expenditures and non-tax incentives in a structured, transparent, and sustainable manner. The ultimate goal is to establish a robust digital platform that consolidates data from multiple sources, estimates revenue foregone with high accuracy, and supports evidence-based policymaking and tax reform.

Deliverable 3 delivered a fully functional system with user interface, backend logic, core features, and reporting capabilities. Deliverable 4 concentrates on refinement based on stakeholder feedback and rigorous testing of tax expenditure calculations — the core function (“heart”) of TaxETS.

3 Key Schedule Change

The original plan for Deliverable 4 envisaged comprehensive testing and final validation of tax expenditure calculations across all tax types (Profit Tax, Personal Income Tax, VAT, Excise Tax, Customs Duty, Import/Domestic VAT, and Non-Tax incentives) within the initially agreed timeline.

During implementation, it became evident that achieving the full scope of testing with the desired level of accuracy and completeness would require additional time. Profit Tax proved particularly challenging due to its complexity and the need to integrate data from multiple sources (MOIC, MPI, MOLSW, SEZOs, Lao Securities Exchange, TaxRIS, etc.) while applying numerous interdependent legal provisions. Similar data availability constraints exist for other tax types — for example, Personal Income Tax calculations currently rely on assumptions derived directly from the law, as Lao PDR does not yet issue personal TINs.

Recognising these realities early, and in close consultation with the Fiscal Policy Department, the Ministry of Finance Expert(s), and UNDP, we proposed — and received approval for — a pragmatic schedule adjustment. This involved:

- Prioritising full development and exhaustive testing of the most complex module (Profit Tax) to establish a robust foundation and valuable lessons for the remaining taxes.
- Simultaneously preparing complete import templates, calculation logic, and reporting functions for all other tax types using the prototypes and sample data generously provided by the MOF Expert.
- Deferring final end-to-end testing with large/real datasets for the remaining taxes to the training/handover phase, when additional data and expert input will be readily available.

This focused approach necessitated a short extension to the overall project timeline, which was formally requested and granted. The adjustment has delivered outstanding results: the system now fully covers all tax types with production-ready functionality. Profit Tax TE calculations have been validated as 100 % accurate, and the remaining modules have been successfully tested against the MOF Expert's sample calculations and are immediately operational.

We are grateful to FPD, the MOF Expert(s), and UNDP for their understanding and flexibility, which enabled us to maintain the highest quality standards while ensuring the system is comprehensive, accurate, and future-proof.

4 Improvement Made according to feedback from UNDP

We received detailed, constructive, and extremely helpful feedback from UNDP on 7 July 2025 (full text in Appendix 1). We are pleased to confirm that virtually all suggestions have been implemented or are in the final stage of implementation. The only remaining item — full MFA and HTTPS deployment — awaits the registration of an official domain (e.g., taxets.mof.gov.la) and SSL certificate, which is currently under discussion between FPD and the MoF Technology Department. APIS remains ready to assist immediately once the domain is approved.

Key implemented improvements include:

1. User Management

- **Authentication (MFA Implementation):** UNDP suggested implementing Multi-Factor Authentication (MFA) due to sensitive data.
 - APIS Response: Data is currently plain text during development; plan to switch to encrypted mode post-completion (version 3.0 or 4.0). Proposal: Encrypt business/company text data, keep numeric/less sensitive data plain text to avoid system overload.
- **Access Security (IP and SSL):** UNDP noted temporary IP-based access.
 - APIS Response: IP is interim; SSL certificate to be applied after official domain registration. Suggest registering a domain and SSL (small annual fee); APIS can handle if authorized.
- **Email Verification:** UNDP highlighted ability to use unverified emails (e.g., test1@gmail.com).
 - APIS Response: Will add email verification.
 - Status: In progress (to be updated post-SSL if required).

- **Password Strength:** UNDP noted weak passwords (e.g., '123456') are allowed; recommend letters and symbols.
 - APIS Response: Will enforce strong passwords (uppercase, numbers, special characters).
- **User Creation Restrictions:** UNDP suggested limiting to Super Admin only.
 - APIS Response: Managed via dynamic roles; can revoke rights easily. Roles will be redefined pre-live (post-version 3.0/4.0) to fit MOF/UNDP structure.

2. Role Management

- **Guidelines and Manuals:** UNDP requested clear guidelines on create/read/update/delete functions and review of manuals.
 - APIS Response: Will update user manual with more comprehensive content.

3. Menu: Get Tax Data by Importing from Excel and by API

- **Function Accessibility:** UNDP observed functions fail for non-Super Admin accounts; assume Super Admin-only.
 - APIS Response: Access is role-based; adjust roles or create new ones for permissions. Update user docs for clarity.
- **Visibility for Unauthorized Users:** UNDP suggested hiding functions for non-designated accounts.
 - APIS Response: Will consider hiding or notifying unauthorized access.

4. Menu: Get Tax Data by Import from Excel -> Data Requirements to Estimate TE -> Profit Tax

- **Edit/Delete Options and TIN Updates:** UNDP noted lack of edit/delete for lines/batches; TIN may need updates.
 - APIS Response: Options exist but hidden (scroll right); may hide columns or adjust in future versions based on user requests.

5. User Activity Tracking

- **Tracking Edits/Deletions/Submissions:** UNDP requested logging for security and recovery.
 - APIS Response: Currently access logs only; will expand in version 3.0 with specifics (e.g., sensitive data operations). Need details on what to track to avoid resource overload. Can will log with Menu: System -> Operation Logs).

6. Admin Tracking and Monitoring

- **Error/User Logs and Export:** UNDP requested error log viewer, user transaction logs (for security/undo), and export to CSV/PDF.
 - APIS Response: Will update in version 3.0; need specifics on transaction types/data to track (e.g., avoid logging all API imports to save resources). Can will log with Menu: System -> Operation Logs).

7. Backup and Restore

- **For Accidental Deletions:** UNDP suggested features for data recovery.
 - APIS Response: Will add in version 3.0/4.0: Auto backups (e.g., midnight/2-3am, configurable); manual ad-hoc via menu (recommend off-peak to avoid slowdowns). Daily backup is made daily at 2am, overwritten after 30 days; directory: /var/backups/mysql. UI function via Menu: System -> Backup/Restore; configurable, with role-based limits.
- **Data Sustainability for Growth:** UNDP raised concerns about performance with increasing data.
 - APIS Response: Optimization requires resources (human/hardware); support during 12-month warranty. Post-warranty: Recommend maintenance contract renewal, hiring external help, or training MOF staff. Open-source nature allows flexibility but needs expertise.

Almost all of feedback and our response with some proposal have been implemented, except 1 issue. Aligning with UNDP feedback to implement MFA for secure access by users to sensitive TaxETS data. To implement this, e.g. to configure the system for HTTPS access, we need to register an official domain name for TaxETS (e.g., taxets.mof.gov.la and similar) and an SSL certificate and then from a Lao service provider, to be noted that this will be involving a small annual fee. We have raise this issue to FPD and it is being considered and in discussion with Technology Department of MOF.

These actions are critical to finalize before the planned handover by September 2025, ensuring a secure and robust TaxETS system.

5 Tax Expenditure Calculation Testing

Testing of tax expenditure calculations was conducted in very close collaboration with the Ministry of Finance Expert(s), whose prototypes, sample datasets, legal analysis, and continuous feedback were indispensable to achieving the current level of accuracy.

5.1 Profit Tax Testing

The MOF Expert kindly provided a carefully designed test dataset of 21 records that covers the majority of real-world scenarios. We split this dataset into the five source-specific files required by TaxETS architecture (MOIC, MOLSW, SEZO, Lao Securities Exchange, and TaxRIS/Profit Tax file). After import, the system reproduced the Expert's Excel results with 100 % accuracy in all tested cases.

Multiple live online sessions were held with the Expert during which we jointly modified parameters (number of employees, tax year, investment promotion status, etc.). In every instance, TaxETS produced identical results to the Expert's manual calculations. This gives us full confidence in the Profit Tax module.

We extend our deepest appreciation to the MOF Expert for the excellent test file and patient guidance throughout this complex phase.

5.1.1 Data Structure of Test File provided by Expert

Data structure of MS Excel testing file provided by the Expert is shown below, full file presentation is shown in Appendix 2. It is to be noted that the presentation below shows how data are provided, where some columns are direct input and some fields are calculated or derived from the values of another columns:

No.	Column Name	Description	Calculation
A	Year	The fiscal or reporting year for the enterprise's data.	Not calculated; input value.
B	Investment License date	The date when the investment license was issued.	Not calculated; input value.
C	Name of enterprise / branch representative	The official name of the enterprise or its branch representative.	Not calculated; input value.
D	TIN	Tax Identification Number of the enterprise.	Not calculated; input value.
E	Province	The province where the enterprise is located.	Not calculated; input value.
F	District	The district within the province where the enterprise operates.	Not calculated; input value.
G	Zone 1	Indicator for whether the enterprise is in Zone 1	Binary Flag: 1 for yes, 0 for no
H	Zone 2	Indicator for whether the enterprise is in Zone 2	Binary Flag: 1 for yes, 0 for no
I	Zone 3	Indicator for whether the enterprise is in Zone 3	Binary Flag: 1 for yes, 0 for no
J	Sector	The industry sector of the enterprise (e.g., Agriculture, Banking, Commerce).	Not calculated; input value.
K	Revenue	Total revenue generated by the enterprise in the reporting period (in Kip).	Not calculated; input value.
L	Expense	Total expenses incurred by the enterprise in the reporting period (in Kip).	Not calculated; input value.
M	Net profit	The net profit or loss for the enterprise.	Calculated as Revenue - Expense.

N	Re-invest net profit	The portion of net profit that is re-invested (in Kip).	Not calculated; input value (empty if not applicable).
O	PT paid	Profit Tax actually paid by the enterprise (in Kip).	Not calculated; input value.
P	Benchmark PT Rate	The standard or benchmark profit tax rate applicable (e.g., 0.2 or 0.24).	Calculated as 0.2 if reporting year is 2022, 0.35 if sector is mining, if not mining then 0.24 if before 2020 otherwise 0.2
Q	Benchmark PT	The benchmark profit tax amount based on net profit and rate.	Calculated as Net profit * Benchmark PT Rate (only if net profit is positive; otherwise empty or zero).
R	PT TE	Profit Tax TE, representing the forgone tax revenue due to exemptions or incentives.	Calculated as Benchmark PT - PT paid (if positive; otherwise empty or zero).
S	Tax holiday period (years)	The duration of the tax holiday granted to the enterprise in years.	Not calculated; input value
T	Date of Registration License	The date of registration license,	Not calculated; input value
U	Businesses engaged in human resource development	Indicator for enterprises involved in human resource development	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
V	Businesses that apply innovative technologies that are environmentally friendly, resource-efficient, and use clean energy for production	Indicator for enterprises using innovative, eco-friendly technologies	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
W	SEZ developer	Indicator for Special Economic Zone (SEZ) developers	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
Z	SEZ investor	Indicator for investors in Special Economic Zones	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
Y	Activities on production industry, services on tourism industry development, services in public	Indicator for enterprises in specified production, tourism, health, education, sport, or real estate	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives

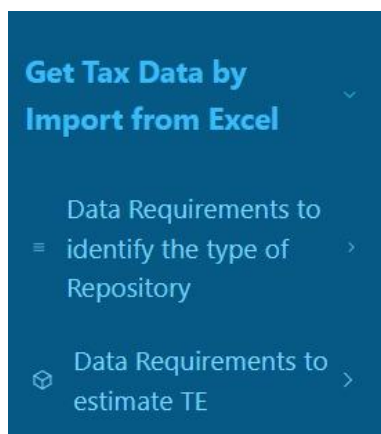
	health, education, sport and physical activity, and real estate development	activities	
Z	Income from activities that provide a public benefit or social purpose, such as art, sport etc	Indicator for income from public benefit activities like art or sport	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AA	Rent from the assets of a business operator who complies with their income tax obligations	Indicator for rental income from compliant business operators	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AB	Income from the transfer of real estate rights recorded in the balance sheet of a business operator	Indicator for income from real estate rights transfers	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AC	Activity 1 of Art. 9, IPL	Indicator for Activity 1 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AD	Activity 2 of Art. 9, IPL	Indicator for Activity 2 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AE	Activity 3 of Art. 9, IPL	Indicator for Activity 3 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AF	Activity 4 of Art. 9, IPL	Indicator for Activity 4 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AG	Activity 5 of Art. 9, IPL	Indicator for Activity 5 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AH	Activity 6 of Art. 9, IPL	Indicator for Activity 6 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AI	Activity 7 of Art. 9, IPL	Indicator for Activity 7 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AJ	Activity 8 of Art. 9, IPL	Indicator for Activity 8 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives
AK	Activity 9 of Art. 9, IPL	Indicator for Activity 9 under Article 9 of the Investment Promotion Law	Binary Flag: 1 for yes, blank for no; qualifies for tax incentives

AL	List of enterprises holding VAT system	Indicator or value for enterprises registered in the VAT system	Binary Flag: 1 for yes, blank for no;
AM	date of re-invest	The date when net profit was re-invested.	Not calculated; input value
AN	Amount of net-profit re-invested (billion Kip)	The amount of net profit re-invested, in billion Kip.	Not calculated; input value
AO	Amount of total assets (billion Kip)	Total assets of the enterprise, in billion Kip	Not calculated; input value
AP	Annual turnover (billion Kip)	Annual sales or turnover, in billion Kip.	Not calculated; input value
AQ	Numbers of staff	The number of employees or staff.	Not calculated; input value
AR	Registration date of firms in Lao Stock Exchange	The date of registration on the Lao Stock Exchange.	Not calculated; input value
AS	TE#1	TE category #1 indicator .	Not calculated; input value. code/ value identifying the type of TE
AT	TE#2	TE category #2 indicator .	Not calculated; input value. code/ value identifying the type of TE.
AU	TE#3	TE category #3 indicator .	Not calculated; input value. code/ value identifying the type of TE.
AV	TE#4	TE category #4 indicator .	Not calculated; input value. code/ value identifying the type of TE.
AW	TE#5	TE category #5 indicator .	Not calculated; input value. code/ value identifying the type of TE.
AX	TE#6	TE category #6 indicator .	Not calculated; input value. code/ value identifying the type of TE.
AY	TE#7	TE category #7 indicator .	Not calculated; input value. code/ value identifying the type of TE.
AZ	TE#8	TE category #8 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BA	TE#9	TE category #9 indicator .	Not calculated; input value. code/

			value identifying the type of TE.
BB	TE#10	TE category #10 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BC	TE#11	TE category #11 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BD	TE#12	TE category #12 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BE	TE#13	TE category #13 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BF	TE#14	TE category #14 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BG	TE#15	TE category #15 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BH	TE#16	TE category #16 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BI	TE#17	TE category #17 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BJ	TE#18	TE category #18 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BK	TE#19	TE category #19 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BL	TE#20	TE category #20 indicator .	Not calculated; input value. code/ value identifying the type of TE.
BM	TE#Other	TE category for other or miscellaneous types (possibly a code or value; e.g., "Other" if applicable).	Not calculated; input value. code/ value identifying the type of TE.
BN	TE#	Summary from AS (TE#1) to BM (TE#other)	Show what TE# is/are applied
BN	Profit Tax TE	The total Profit Tax TE amount (in Kip), summarizing the forgone revenue.	Calculated as Benchmark PT - PT paid (same as column R)

5.1.2 Import Data Rearrangement

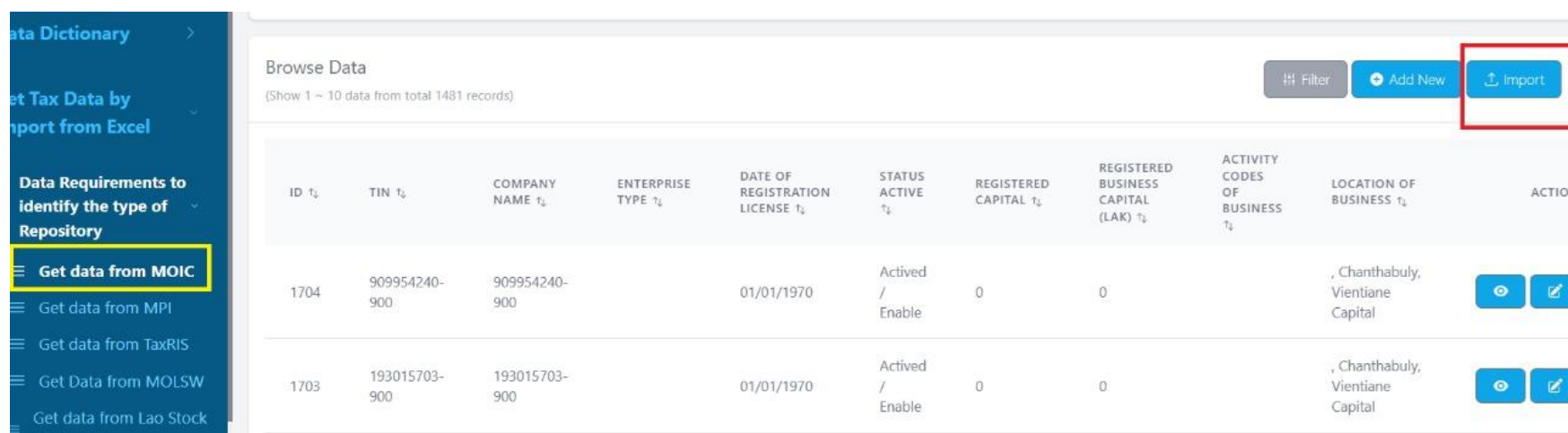
According to our system design, we categorize data to be linked or imported from external sources in 2 main groups:



collected from many different sources, most of which only use the same enterprise identification number (TIN) used as the key to link.

Therefore, the data import process will also have to use many functions at the same time, corresponding to each type of data to be imported into the system.

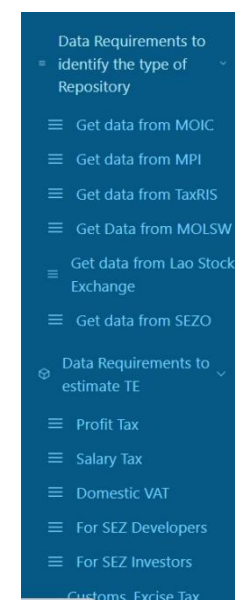
For example, data from MOIC related to Enterprise information will be imported from a separate file that can be expected to be received from MOIC. The file will have to be standardized (correct data type, and only store values, not formulas...) before importing into the system. Use this function to import the correct file format that you can download from the system.



Browse Data
(Show 1 ~ 10 data from total 1481 records)

ID	TIN	COMPANY NAME	ENTERPRISE TYPE	DATE OF REGISTRATION LICENSE	STATUS ACTIVE	REGISTERED CAPITAL	REGISTERED BUSINESS CAPITAL (LAK)	ACTIVITY CODES OF BUSINESS	LOCATION OF BUSINESS	ACTION
1704	909954240-900	909954240-900		01/01/1970	Activated / Enable	0	0		, Chanthabuly, Vientiane Capital	 
1703	193015703-900	193015703-900		01/01/1970	Activated / Enable	0	0		, Chanthabuly, Vientiane Capital	 

All other data types are implemented in a similar way (MPI, TaxRIS, MOLSW, Lao Stock Exchange...)



And then the most important type of data related to calculating Profit taxes.

Data Dictionary

Get Tax Data by Import from Excel

Data Requirements to identify the type of Repository

Data Requirements to estimate TE

- Profit Tax**
- Salary Tax
- Domestic VAT
- For SEZ Developers

Browse Data
(Show 1 - 10 data from total 333 records)

Search...

Add New **Import**

TIN	FILING TYPE	FILING PERIOD	INPUT DATE	YEAR	REVENUE	EXPENDITURE	TOTAL PROFIT	PROFIT TAX RATE	AMOUNT OF PROFIT TAX COLLECTED	TOTAL ASSE
232280925-900			2025-08-06	2018	1,098,415,500.00	1,042,630,615.00	55,784,885.00	24%	13,200,000.00	0.00
773663486-000			2025-08-06	2018	12,000,000.00	0.00	12,000,000.00	24%	1,800,000.00	0.00
057258273-000			2025-08-06	2018	4,280,799,900.00	3,760,000,000.00	520,799,900.00	24%	120,000,000.00	0.00
822868467...			2025...							

So, from 1 file from the expert, we have split into 5 files the temple of which will be used for the import from MS Excel File in the future:

Now from this file we need to split in to 5 files:

- File for import as Data Import from MOIC
- File for import as Data Import from MOLSW
- File for import as Data Import from SEZO
- File for import as Data Import from Lao Stock Exchange
- File for import for estimate Profit Tax (Should be in the TaxRIS)

Below is the structure of each file, Data presentation of each files can be found in Appendix 3.

5.1.2.1 Data Import from MOIC (Extracted from Expert's MS Excel file)

Excel Column	Column Name	Description
A	TIN	Tax Identification Number of the enterprise.

B	Company_Name	The official name of the company or enterprise.
C	ProjectID	Unique identifier for the project associated with the enterprise.
D	Province	Code or name of the province where the enterprise is located.
E	District	Code or name of the district within the province.
F	Village	Name of the village or specific location within the district.
G	Enterprise_type	Type or classification of the enterprise (e.g., limited company).
H	Date_of_establishment_Investment_License	Date of establishment or issuance of the investment license.
I	Date_have_revenue	Date when the enterprise first generated revenue.
J	Date_grant_incentives	Date when tax or other incentives were granted.
K	Incentives_tax_policy	Description or code for the tax incentives policy applied.
L	Investors_fund_rate	Rate or percentage of funds provided by investors.
M	StatusActive	Status indicating if the enterprise is active (e.g., 1 for active).
N	Business_size_by_staff	Size of the business categorized by number of staff.
O	Registered_business_capital_LAK	Registered capital of the business in Lao Kip (LAK).
P	Human_resource_development_scope	Indicator for scope involving human resource development (e.g., 1 if applicable, used for incentives).
Q	Apply_innovative_technologies_scope	Indicator for scope applying innovative, environmentally friendly technologies (e.g., 1 if applicable, used for incentives).
R	Belong_points_2_of_Art_9	Indicator if the enterprise belongs to point 2 of Article 9 (likely from Investment Promotion Law; 1 if yes, used for incentives).
S	Belong_points_3_of_Art_9	Indicator if the enterprise belongs to point 3 of Article 9 (likely from Investment Promotion Law; 1 if yes, used for incentives).
T	Belong_points_4_of_Art_9	Indicator if the enterprise belongs to point 4 of Article 9 (likely from Investment Promotion Law; 1 if yes, used for incentives).
U	Belong_points_5_of_Art_9	Indicator if the enterprise belongs to point 5 of Article 9 (likely from Investment Promotion Law; 1 if yes, used for incentives).

V	Belong_points_6_of_Art_9	Indicator if the enterprise belongs to point 6 of Article 9 (likely from Investment Promotion Law; 1 if yes, used for incentives).
W	Production_industry_scope	Indicator for scope in production industry activities (e.g., 1 if applicable).
X	Services_on_tourism_industry_development_scope	Indicator for scope in services for tourism industry development (e.g., 1 if applicable).
Y	Services_in_public_health_scope	Indicator for scope in public health services (e.g., 1 if applicable).
Z	Education_scope	Indicator for scope in education services (e.g., 1 if applicable).
AA	Sport_and_physical_activity_scope	Indicator for scope in sport and physical activity services (e.g., 1 if applicable).
AB	Real_estate_development_scope	Indicator for scope in real estate development (e.g., 1 if applicable).
AC	Micro_enterprises_scope	Indicator for micro-enterprise scope (e.g., 1 if applicable).
AD	Agriculture_and_handicraft	Indicator or category for agriculture and handicraft sector (e.g., 1 if applicable).
AE	Industry_and_manufacturing	Indicator or category for industry and manufacturing sector (e.g., 1 if applicable).
AF	Commerce_and_services	Indicator or category for commerce and services sector (e.g., 1 if applicable).
AG	Electric_power_and_minning	Indicator or category for electric power and mining sector (e.g., 1 if applicable).
AH	EmailAddress	Email address associated with the enterprise.
AI	Services_in_public_health_scope	Indicator for scope in public health services (possibly a duplicate or variant of column Y; e.g., 1 if applicable).
AJ	main_category	Main category or sector classification of the enterprise.
AK	Industry_Sub_Category	Sub-category within the main industry or sector

5.1.2.2 Data Import from MOLSW (Extracted from Expert's MS Excel file)

Excel Column	Column Name	Description
A	TIN	Tax Identification Number of the enterprise.
B	Numbers_staff	The number of staff or employees in the enterprise for the specified year.
C	Year_data	The year to which the data (e.g., number of staff) pertains.

5.1.2.3 Data Import from SEZO (Extracted from Expert's MS Excel file)

Excel Column	Column Name	Description
A	TIN	Tax Identification Number of the enterprise.
B	Date_Permission_License	The date of permission or license issuance, likely in serial date format (e.g., 42267 represents a specific date).
C	ProvinceID	Identifier or code for the province where the enterprise is located.
D	DistrictID	Identifier or code for the district within the province.
E	Sector	Code or identifier for the industry sector of the enterprise (e.g., 6 possibly for Consultancy or similar).
F	Developer	Indicator for whether the enterprise is an SEZ (Special Economic Zone) developer (e.g., 1 for yes, blank for no).
G	Investors	Indicator for whether the enterprise is an SEZ investor (e.g., 1 for yes, blank for no).
H	amount_of_construction_road_electricity_water_wastewater	Amount spent on construction of roads, electricity, water, and wastewater infrastructure (in unspecified currency; may be empty).
I	amount_of_construction_of_any_other_infrastructure	Amount spent on construction of any other infrastructure (in unspecified currency; may be empty).
J	amount_of_the_use_electricity_water_in_production	Amount for the use of electricity and water in production (in

		unspecified currency; may be empty).
K	amount_of_construction_development_of_infrastructures	Amount for construction and development of infrastructures (in unspecified currency; may be empty).

5.1.2.4 Data Import from Lao Stock Exchange (Extracted from Expert's MS Excel file)

Excel Column	Column Name	Description
B	TIN	Tax Identification Number of the enterprise.
C	Date_join	Date when the enterprise joined the Lao Stock Exchange (e.g., in format YYYY-DD-MM; may be empty if not joined).
D	Status_join	Status indicating whether the enterprise has joined the stock exchange (e.g., 1 for joined, 0 for not joined).
E	UserUpdate	Identifier or name of the user who last updated the record (may be empty if no update recorded).

5.1.2.5 File for import for estimate Profit Tax (Extracted from Expert's MS Excel file)

Excel Column	Column Name	Description
A	TIN	Tax Identification Number of the enterprise.
B	Filing_type	Type of tax filing submitted by the enterprise (e.g., annual, quarterly; may be empty if not specified).
C	Filing_period	The period covered by the tax filing (e.g., month, quarter; may be empty if not specified).
D	Input_date	Date when the data was input or recorded in the system (may be empty if not specified).
E	Year	The fiscal or reporting year for the profit tax data.
F	Revenue	Total revenue generated by the enterprise in the reporting year (in the specified currency).
G	Expenditure	Total expenditures or expenses incurred by the enterprise in the reporting year (in the specified currency).
H	Total_Profit	The total profit or loss calculated as Revenue minus Expenditure (in the specified

		currency).
I	Profit_tax_rate	The applicable profit tax rate for the enterprise (e.g., 0.2 or 0.24 or 0.35).
J	Amount_of_profit_tax_collected	The amount of profit tax actually collected or paid by the enterprise (in the specified currency; may be empty or zero if no tax due).
K	total_assets	Total assets of the enterprise as reported (in the specified currency; may be empty).
L	Currency	The currency in which the financial amounts are reported (e.g., LAK for Lao Kip).

It is to be noted there is still an item (column) to be set before we can make calculation of Profit Tax TE, the Date that company start having revenue, to do this we have agreed with expert to assume that the date of start having revenue is just the starting of year that we make TE estimation or at least before this year.

By this way we can see the result of PT TE match perfectly with the Result calculated in the expert MS Excel file. We have also organized few online testing sessions with Expert where we together manipulate some parameter like number of staff, year of tax calculation, etc. and we have confirmed that the outcomes match each other perfectly.

Below is an example:

For a company with TIN: 763723844-000, PT TE is exactly the same as in the MS Excel from Expert, here is the screenshot from TaxETS:

ID	TIN	TE PROFIT PROVISION	TE RATE	BENCHMARK VALUE	PT PAID	PT TE	TE YEAR	TE FROM DATE	TE TO DATE	DATETIME CREATED	ACTION
4674	763723844-000	1000	20%	7,800,000.00	2,500,000.00	5,300,000.00	2020	2020-01-01 00:00:00	2025-10-07 13:30:10	2025-10-07 13:30:10	

And here is MS Excel File from Expert:

Year	Investment License date	TIN	Sector	Revenue	Expense	Net profit	Re-invest net profit	PT paid	Benchmark PT Rate	Benchmark PT	PT TE	Tax holiday period (years)	Date of Registration License	Budgetary allocation
2021	17-Sep-14	763723844-000	ulture	200,000,000	80,000,000	120,000,000		2,000,000	20%	24,000,000.00	22,000,000.00		1-Jan-23	
2020	17-Sep-17	763723844-000	ulture	128,000,000	89,000,000	39,000,000		2,500,000	20%	7,800,000.00	5,300,000.00			

To be noted that we need to change the number of staff to be more than 99 person, since in reality according to TE Profit Provision is No. 1: ITL Art. (amended) 16(2), Rate Relief, A newly registered small entity (**6-50 staff**, total assets not exceeding LAK1.5 billion, & annual turnover $\leq$ LAK3 billion ^{a)} is subject to a profit tax rate of **3%** for 3 years, after which the standard PT rate applies), and according to TE Profit Provision No. 2: A newly registered medium-sized entity (**51-99 staff**, total assets not exceeding LAK6 billion, & annual turnover $\leq$ LAK6 billion) is subject to a profit tax rate of **5%** for 3 years, after which the standard PT rate applies).

We have also manipulate different parameter such as Year and other, the result shows that our application makes correct PT TE Calculation, and taking into account all parameter (variable) to identify Profit Tax TE Rate including those written Benchmark and different kind of provision.

5.2 Personal Income Tax, VAT, Excise Tax, Customs Duty, and Non-Tax Incentives

For the remaining taxes, we again worked hand-in-hand with the MOF Expert to:

- analyse the calculation prototypes and legal provisions provided,
- define minimum data requirements,
- create standardised import templates (downloadable from the system),
- implement calculation logic that faithfully follow the Expert's logic and the relevant laws,
- successfully validate results against the Expert's sample calculations (examples shown below).

The testing results for Personal Income Tax, VAT, Excise Tax, and Customs Duty all match the Expert's Excel calculations exactly (screenshots provided). Any minor discrepancies in template structure versus the Expert files were resolved by adding flexible classification fields (e.g., "Type of goods/service", HS Code mapping) so that the system remains compatible with both current simplified data and future detailed data from TaxRIS/ASYCUDA.

We are extremely grateful to the MOF Expert for the provision of prototypes, sample files, legal references, and prompt responses, which enabled us to complete the logic for all remaining taxes within the adjusted timeline.

For these 4 types of taxes, the big difference compared to Profit Tax is that the tax calculation cycle is shorter, almost as if there is no tax calculation cycle. Because this type of tax can arise at any time during the year or may not arise this year but will arise next year. Therefore, we build a data storage and processing system based on the event principle (that is, only need to pay attention to the time when this tax transaction arises), then, if it is necessary to include this tax transaction in the tax calculation cycle in the related reports, we will use the tax calculation cycle in the style of "from date,... to date". This flexible application will help manage and calculate all types of taxes regardless of the period they arise. Of course, we always must ensure the principle of data consistency, and that data is unique.

Profit tax has an annual cycle, while these 4 types of taxes can arise at any time of the year or day. In a day, a business or individual (corresponding to TIN code - if any) can incur any taxes, multiple times, with the same type of goods/products or revenue sources. The processing of this input data duplication mainly relies on related tax management systems (TaxRIS, Asycuda...). Because our system is designed to have a long-term goal of receiving tax data through APIs, manual data entry only occurs in the functional testing or design phase. During the operational process, standardized/encrypted/duplicated data will be received from external sources. Here, the function of importing data from Excel file will also be designed according to the same principle (it will be considered that for each type of tax, a tax will be identified as unique by the transaction code), like a currency transaction in the banking system.

What does this mean? It means that in practice, a business/individual (represented by TIN) may be subject to multiple taxes at the same time or very close together. For example, an individual may sell multiple houses on the same day, or even at the same time, and therefore, he may have to pay personal income tax multiple times (or once) for each house sold. The system will have to track and store each of those taxes (if any) in separate data records. Or another example is that an individual may sell goods online with multiple orders at the same time. Each of those orders will have information to specifically identify tax activities such as transaction code/invoice code, type of goods, invoice amount, corresponding tax amount (if any). Even within the same invoice, there will have to be classification elements to identify separate taxes with different specific tax rates.

Based on the formula and reserve classification we received from the sample provided by the Ministry of Finance expert, for these 4 taxes, the minimum information we need to operate (import data / Run calculate Benchmark / build Result report) the TaxETS system includes:

Item	Sub Item (if have)	Description	Note
1. TIN		is the tax code of an enterprise/individual in the national tax management system	Mandatory
2. Datetime of Tax Events		The time an Expenditure (tax)	Mandatory

		occurs	
3. Total Tax Amount / Invoice		Total Amount of goods/services ... to be taxed	Mandatory
4. Classify / Estimate/ Matching Tax Rate	4. 1. Category of the Goods (variable name/ HS code...)	Information to classify goods/services ... of the tax. This information is often built in a way that inherits existing external data systems (otherwise it would have to be used by setting them as variables). For example, HS Code in customs system....	Mandatory. The information used to classify goods and services... in the 4 tax types is different. Therefore, each type will have a specific classification variable name. Please see the description table below for each tax type. In MS Excel template Import files.
	4.2. Tax Rate	Tax rates payable correspond to the list of classified goods.	
	4.3. Rule/Exception (if have)	The reference rule/reference document/exception rule (if any) serves as the legal basis for calculating tax and determining reporting items/provisions. Ex: 21 - Exemption - ITL., Art. 35.3	
5. Tax PAID/ Collected		Corresponding tax amount paid	Optional (if have)
6. Tax Benchmark/TE		Benchmark tax amount	The system will automatically calculate this value according to the prescribed rules and tax rates.
7. TE Value remaing (TE-PAID)	Tax Bench - PAID	Value / Remaining tax payable	The system will automatically calculate this value according to the prescribed rules and tax rates.
8. Provisions (if have)		Identify the corresponding Provision according to the list provided by the MOF expert	The system will automatically assign this value according to the prescribed rules and tax rates.

We need to fill in minimum data for the first 4 items (numbers 1 to 4) to enter data for all taxes. The remaining information is not mandatory and can be temporarily ignored as they are only used for future reporting or record keeping purposes.

Our biggest challenge was to categorize the spending taxes and then figure out the tax rates for these taxes. It is Item number 4 " Classify / Estimate/ Matching Tax Rate ".

We have designed the data for each tax type, and built in data entry functions (manual add, edit, delete and use MS Excel files), standard tax calculation functions and reporting functions. Below, we will present and explain the relevant contents for each tax type.

The general approach is:

1. Review sample exemption/deduction provisions... according to legal documents and other exceptional documents/regulations/clauses or the experience of the Ministry of Finance expert requested/recommended.
2. Review/matching the minimum data format required for each tax.
3. Inherit data content from MS Excel files provided by MOF experts: Analyze what information will be used and how it will be normalized into labeled and meaningful variables; what information will be temporarily unused (or will have to wait until new additional data becomes available before it is truly useful/meaningful).
4. Create data templates and build standard tax calculation functions, build reporting functions

5.2.1 5.2.1. Personal Tax - Individual Tax

5.2.1.1 Individual calculation rule prototype and legal documents

- **Legal documents:**
 - + *Investment Law 2016_English version.pdf*
 - + *List of Investment Promotion Activity (English version)*
- **Provision list provided by MOF expert (from MS Excel file)**

TE #	Type	Legal Basis	TE Description
21	Exemption	ITL., Art. 35.3	Additional working hours or overtime payments only for employees receiving a basic salary of LAK2 million or less (TE)
22	Exemption	ITL., Art. 35.20	Uniforms and equipment for preventing labour accidents (TE)
23	Exemption	ITL., Art.35.3	Personal allowance for spouse, children aged below 18 (LAK5m up to a maximum of LAK15m per year)
24	Exemption	ITL., Art.35.3	Allowance for government personnel, allowance for national assembly member and provincial people's assembly
25	Exemption	ITL., Art.35.3	Allowance for students
26	Exemption	ITL., Art.35.4	Profits derived from the sale of shares from a company listed on the Lao stock exchange.
27	Exemption	ITL., Art.35.4	Dividends distributed to partners or shareholders of a company registered on the stock exchange
28	Exemption	ITL., Art.35.11	Interest on deposits in the banking system, Income from bonds issued by listed companies and government bonds
29	Exemption	ITL., Art.35.13	Bonus or gift paid by government to a person with outstanding performance in monitoring, searching, protecting, prevention or blockage of illegal acts

30	Deduction	Law on Social Security Art 74.2;	Employee (5.5%) social security contributions are deductible from individual income tax. The contributions are computed on a maximum base of LAK4.5m per month; therefore, the maximum amount deductible is LAK247,500 (employee).
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Table 5.2.1.1

- Individual Tax Rate Summary Table

Tax Unit	Tax Base	Tax Period	Tax Rate						
			Employment Income			Financial Income		Other Income	
			Monthly income (LAK)		Marginal PIT rate				
Individual	(i) Employment Income, (ii) Financial Income and (iii) Other income	Monthly and periodically basis	0	1,300,000	0	The sale or transfer of shares of an individual or legal entity	2%	The sale or transfer of agricultural land tenure for agricultural purposes	1%
			1,300,001	5,000,000	5%			Online sales	2%
								Lottery winnings	5%
								Income from rental	10%
			5,000,001	15,000,000	10%	Income from dividends or other benefits to shares or shareholders	10%	Income from sport and performance activities	10%
			15,000,001	25,000,000	15%			Income from non-commercial activities of government and social organizations	10%
								Income from intellectual property	5%
			25,000,001	65,000,000	20%	Loan interest from outside the banking system, income from guarantee	10%	Transfer of real estate rights	2%
			65,000,001	+	25%			Inheritance beyond horizontal lineage	2%
								Consulting fees	5%
								Gift as cash or in kind	5%

Table 5.2.1.2

The above data table applies to periods after 31 December 2023.

We also have an additional provision for the tax rate data table that applies to all personal taxes from before 31 December 2019 as below:

Tax Unit	Tax Base	Tax Period	Employment Income		
			Monthly income (LAK)		Marginal P/T rate
Individual	(i) Employment Income, (ii) Financial Income and (iii) Other income	Monthly and periodically basis	0	1,000,000	0
			1,000,001	3,000,000	5%
			3,000,001	6,000,000	10%
			6,000,001	12,000,000	12%
			12,000,001	24,000,000	15%
			24,000,001	40,000,000	20%
			40,000,001	+	24%

Table 5.2.1.2_addition

5.2.1.2 Define the Minimum Data table

After careful consideration of each case and with an eye to future expansion, we have determined that the minimum data sheet (including both mandatory and optional information) should include at least the following information:

No.	Column Name	Description	Calculation/Input/Remark
1	Uniq ID for all Record in system	ID Transactions in system	Mandatory, Unique Identify
2	TIN of Organization / person	Tax Identification Number of the enterprise.	Mandatory
3	Te Individual Provision (if have)	Provision of this Tax transaction	Option Only for double check
4	Date time appear Income (date time sale/buy...)	The time of transaction of goods/services that may generate tax.	Mandatory
5	Total Individual income (Invoice Amount LAK)	Total Amount of income ... to be taxed	Mandatory
6	Tax rate (if have)	Tax rate	Option Only for double check
7	Individual TE value (if have for compare)	TE Value from other source...	Option Only for double check
8	Currency type/name, default is LAK	Currency	Option
9	Amount of Individual collected/PAID	The amount the taxpayer paid for this tax	Option to Calculate TE Value Remaining
10	TE Individual Tax Benchmark (if have)	TE Value benchmark from other source...	Option Only for double check
11	Variables to classification kind of Individual income tax	Variables are used to classify goods and thereby find the tax rate that will be applied. Please see the table below for details on the values of this variable.	Mandatory. Select a value from the list below to classify the goods/services.... that are subject to this tax.

Table 5.2.1.3

This is also the MS Excel file structure used in the personal/ Individual tax data import function.

Based on the "**Individual Tax Rate Summary Table**" above, which provided by the expert, we have built specific variables, corresponding to the rules / provisions as follows:

With **Employment Income**, two more sub variables need to be added to represent Minimum Monthly Income and Maximum Monthly Income.

ID Type	Type Variable Name	Variables to classification kind of Individual income tax	Effective_date_from	Effective_date_to	Monthly_income_minimum	Monthly_income_maximum	Tax rate	ID_CAT
1.	Individual Income employment	Individual_Income_employment	Unlimited	2019-DEC-31	0	1,000,000	0.00	1
2.	Individual Income employment	Individual_Income_employment	Unlimited	2019-DEC-31	1,000,001	3,000,000	5.00	1
3.	Individual Income employment	Individual_Income_employment	Unlimited	2019-DEC-31	3,000,001	6,000,000	10.00	1
4.	Individual Income employment	Individual_Income_employment	Unlimited	2019-DEC-31	6,000,001	12,000,000	12.00	1
5.	Individual Income employment	Individual_Income_employment	Unlimited	2019-DEC-31	12,000,001	24,000,000	15.00	1
6.	Individual Income employment	Individual_Income_employment	Unlimited	2019-DEC-31	24,000,001	40,000,000	20.00	1
7.	Individual Income employment	Individual_Income_employment	Unlimited	2019-DEC-31	40,000,001	Unlimited	24.00	1
8.	Individual Income employment	Individual_Income_employment	2020-JAN-01	Unlimit	0	1,300,000	0.00	1
9.	Individual Income employment	Individual_Income_employment	2020-JAN-01	Unlimit	1,300,001	5,000,000	5.00	1
10.	Individual Income employment	Individual_Income_employment	2020-JAN-01	Unlimit	5,000,001	15,000,000	10.00	1
11.	Individual Income employment	Individual_Income_employment	2020-JAN-01	Unlimit	15,000,001	25,000,000	15.00	1
12.	Individual Income employment	Individual_Income_employment	2020-JAN-01	Unlimit	25,000,001	65,000,000	20.00	1

13.	Individual Income employment	Individual_Income_employment	2020-JAN-01	Unlimit	65,000,001	Unlimited	25.00	1
14.	The sale or transfer of shares of an individual or legal entity	The_sale_or_transfer_of_shares					2	2
15.	Income from dividends or other benefits to shares or shareholders	Income_from_dividends_or_other_benefits_to_shares					10	2
16.	Loan interest from outside the banking system, income from guarantee	Loan_interest_from_outside_the_banking_system_income					10	2
17.	The sale or transfer of agricultural land tenure for agricultural purposes	The_sale_or_transfer_of_agricultural_land_tenure					1	3
18.	Online sales	Online_sales					2	3
19.	Lottery winnings	Lottery_winnings					5	3
20.	Income from rental	Income_from_rental					10	3
21.	Income from sport and performance activities	Income_from_sport_and_performance_activities					10	3
22.	Income from non-commercial activities of government and social organizations	Income_from_non-commercial_activities_of_government					10	3
23.	Income from intellectual property	Income_from_intellectual_property					5	3
24.	Transfer of real estate rights	Transfer_of_real_estate_rights					2	3
25.	Inheritance beyond horizontal lineage	Inheritance_beyond_horizontal_lineage					2	3
26.	Consulting fees	Consulting_fees					5	3
27.	Gift as cash or in kind	Gift_as_cash_or_in_kind					5	3

Table 5.2.1.4

In which:

+ ID_CAT:

- 1- Employment income
- 2- Financial income
- 3- Other Income

+ Tax_Provision is the ID of the Tax Provision as defined by the expert. The data "91" represents the "Other provision" of the Individual tax, it also has the same meaning as "Undefined" or "Unclassified".

To cover all standard tax calculation rules, we have built a category data table based on the "**Individual Tax Rate Summary Table**" data table provided by experts from the Ministry of Finance. In the future, if it is discovered that new regulations or special provisions affecting tax rates need to be added, this design still allows for flexible expansion without requiring major structural changes.

5.2.1.3 Create template file and build function on Dashboard of TaxETS system.

Below is some functional interface screens related to **Individual income tax**

1. For import and manage Individual income tax

TaxETS

System >

Data Dictionary >

Get Tax Data by Import from Excel >

Data Requirements to identify the type of Repository >

Data Requirements to estimate TE >

- Profit Tax
- Excise Tax Actually
- Custom Tax Actually
- Individual Tax**
- Salary Tax
- Domestic VAT
- For SEZ Developers
- For SEZ Investors

Individual Tax

Dashboard » Individual Tax

Browse Data (Show 1 ~ 9 data from total 9 records)

Search... Add New Import

UNIQ ID FOR ALL RECORD IN SYSTEM	TIN OF PERSON	DATE TIME APPEAR INCOME (DATE TIME SALE/BUY..)	TOTAL VALUE INCOME	TAX RATE (IF HAVE)	INDIVIDUAL TE VALUE FOR COMPARE (IF HAVE)	CURRENCY TYPE/NAME, DEFAULT IS LAK	VARIABLES TO CLASSIFICATION KIND OF INCOME TAX	ACTION
225	202028111-9022	01/01/2019 00:00:00	50,000,000	0.00%	0	LAK	Gift_as_cash_or_in_kind	
2525	12345682	01/01/2023 10:57:00	50,000,000	0.00%	0	LAK	Consulting_fees	
2290	202028111-4545	01/02/2024 00:00:00	5,000,000	0.00%	0	LAK	Consulting_fees	
2288	202028111-4545	01/02/2022 00:00:00	1,000,000	2.00%	0	LAK	Consulting_fees	
345345	123	05/05/2023 00:00:00	8,000,000	0.00%	0	LAK	Inheritance_beyond_horizontal_lineage	

MS Excel template file as described in 5.2.1.2 Define the Minimum Data table. Users can download from the Import function interface.

2. For Run Benchmark Individual Tax

TaxETS

- System >
- Data Dictionary >
- Get Tax Data by Import from Excel >
- Get Tax Data by API >
- Run Tax Expenditure ▾
 - Run Profit Tax Expenditure
 - Run Individual Tax Expenditure**
 - Run Excise Tax Expenditure
 - Run VAT Tax Expenditure
 - Run Custom Tax Expenditure

Run Individual Tax Expenditure

TIN: Start Date *: End Date *:

☐	UNIQ ID FOR ALL RECORD IN SYSTEM	TIN	INPUT DATE	TOTAL INDIVIDUAL INCOME	INDIVIDUAL TAX RATE	TE VALUE	CURRENCY
☐	433	763723844-0121	01/01/2022 07:00:00	1,000,000	2%	23,423	LAK
☐	43334534	763723844-0122	01/01/2022 07:00:00	2,000,000	2%	4,354	LAK
☐	444	763723844-0123	01/01/2022 07:00:00	1,000,000	2%	23,423	LAK
☐	555	763723844-0555	01/02/2022 07:00:00	1,000,000			LAK
☐	345345	123	05/05/2023 07:00:00	8,000,000	0%	0	LAK
☐	2288	202028111-4545	01/02/2022 07:00:00	1,000,000	2%		LAK
☐	2290	202028111-4545	01/02/2024 07:00:00	5,000,000			LAK

Input:

- TIN
- Start date and End date: Date time when you want to start running Benchmark tax calculations for the selected TIN (that means from time - to time)

Output: The system will automatically search for entered tax data that meets the above conditions and calculate the standard tax value (if met).

2. View Result Benchmark Individual Tax

Run Individual Tax Expenditure Result Data												Dashboard » Run Individual Tax Expenditure Result Data
Browse Data												Search...
(Show 1 ~ 1 data from total 1 records)												
UNIQ ID FOR ALL RECORD IN SYSTEM ↑↓	TIN OF PERSON ↑↓	TE INDIVIDUAL PROVISION ↑↓	TOTAL VALUE INCOME ↑↓	TAX RATE ↑↓	INDIVIDUAL TE VALUE FOR COMPARE ↑↓	ចេញបង់ប្រាក់ ពន្ធដារ ↑↓	AMOUNT OF INDIVIDUAL TAX COLLECTED/PAID ↑↓	TE ប្រាក់បង់ ↑↓	DATETIME CREATED ↑↓	USER CREATE/UPDATE ↑↓	NOTE ↑↓	ACTION
345345	123	91	8,000,000	2.00%	160,000	05/05/2023 00:00:00	0	160,000	15/11/2025 21:32:33			
Items per page: 10												

Output:

- + Tax rate
- + Individual TE value for compare (Individual Tax benchmark value)
- + Amount of Individual tax collected/PAID (if have)
- + TE Value (TE = Tax benchmark - Tax PAID)
- + Datetime RUN
- + User ID RUN

5.2.1.4 Individual Tax Test Result.

Here is an example from TaxETS system:

Run Individual Tax Expenditure Result Data Dashboard » Run Individual Tax Expenditure Result Data

Browse Data (Show 1 ~ 1 data from total 1 records)

TIN OF PERSON ↑↓	TE INDIVIDUAL PROVISION ↑↓	TOTAL VALUE INCOME ↑↓	TAX RATE ↑↓	INDIVIDUAL TE VALUE ↑↓	DATE TIME APPEAR ↑↓	AMOUNT OF INDIVIDUAL TAX COLLECTED/PAID ↑↓	INDIVIDUAL TE ↑↓	DATETIME CREATED ↑↓
12345678	21	20,000,000	15.00%	3,000,000	01/01/2019 10:57:00	0	3,000,000	19/11/2025 10:44:58

And now the result from Expert's file:

Date of tax fiing	Name of employee	PTIN	Amount of additional working hours or overtime payments (21)	TE_(21)	Uniforms and equipment for preventing labour accidents (22)	TE_(22)	Amount of personal allowance for spouse (23,1)
1/1/2019	A	12345678	20,000,000.00	3,000,000.00		0.00	

So the TE 1,250,000 is exactly the same.

5.2.2 VAT

5.4.2.1 VAT calculation rule prototype and legal documents

- **Legal documents:**
 - + VAT Law 52 (2014)_English version
 - + VAT Law 48 (2018)_English version
- **Provision list provided by MOF expert (from MS Excel file)**

TE #	Type	Legal Basis	TE Description
------	------	-------------	----------------

31	Exemption	VAT Law Art. 12.1.1	Import of: all types of crop seeds, animals for breeding, animal sperms, vaccines, equipment and liquid nitrogen for storing vaccines and animal sperms, animal feeds, raw materials to produce animal feeds and vaccines;
32	Exemption	VAT Law Art. 12.1.2	Import of: Raw materials used in the production of fertilizers, agroprocessing products, organic fertilizers, [scientific] fertilizers [and] pesticides that are not dangerous to the ecosystem, human and animal health and life
33	Exemption	VAT Law Art. 12.1.3	Import of: Equipment and machinery used in agriculture;
34	Exemption	VAT Law Art. 12.1.5; LOIP Art. 12	Import of: Materials [and] equipment that could not be supplied or produced in Lao PDR and machinery that are used as fixed assets and used directly in production;
35	Exemption	VAT Law Art 12.1.13	Import of: Animal medicines, artificial organs for transplantation for animal bodies
36	Exemption	VAT Law Art 12.1.14	Import of: Traditional medicines, artificial organs for transplantation for human bodies, human blood and supporting equipment for patients, the disabled and elderly;
37	Exemption	VAT Law Art 12.1.15	Import of: Medical tools [and] equipment, [laboratory] equipments or wheelchairs for public services of hospitals, health care centers.
38	Exemption	VAT Law Art 12.1.16	Import of: Vehicles serving professional activities, contributing to public benefits such as fire trucks, ambulances, vehicles equipped with repair facilities, outside television and radio broadcast vehicles and other professional vehicles.
39	Exemption	VAT Law Art. 12.2.1	Domestic supply of: Unprocessed agricultural or preliminary processed products including peeled, grinded [and] milled;
40	Exemption	VAT Law Art. 12.2.2	Domestic supply of: All kinds of live or deceased animals, whole or in parts, that are unprocessed or preliminary processed to ensure freshness or non-perishability;
41	Exemption	VAT Law Art. 12.2.3	Domestic supply of: afforestation, plantation of industrial, fruit and medicinal trees;

42	Exemption	VAT Law Art. 12.2.4	Domestic supply of: all types of crop seeds, animals for breeding, animal feeds, vaccines, raw materials to produce animal feeds and vaccines;
43	Exemption	VAT Law Art. 12.2.5	Domestic supply of: Raw materials used in the production of fertilizers, agroprocessing products, organic fertilizers, [scientific] fertilizers and pesticides that are not dangerous to the ecosystem, human and animal health and life;
44	Exemption	VAT Law Art. 12.2.7	Domestic supply of: equipment and machinery used in agricultural activities;
45	Exemption	VAT Law Art. 12.2.11	Domestic supply of: textbooks and learning materials
46	Exemption	VAT Law Art. 12.2.13	The supply of education services by e.g. childcare centres, kindergartens, primary and secondary schools, vocational schools, colleges, academies, universities, sporting and athletic schools.
47	Exemption	VAT Law Art. 12.2.17	The examination, treatment, and diagnosis of humans and animals
48	Exemption	VAT Law Art. 12.2.18	Animal medicines and artificial organs for transplantation of animal bodies
49	Exemption	VAT Law Art. 12.2.19	Traditional medicines, artificial organs for transplantation of human bodies, human blood, equipment or wheelchairs for patients, disabled and elderly;
50	Exemption	VAT Law Art. 12.2.20	Medical tools, equipment, [laboratory] analytical tools for hospitals, healthcare centers to serve society;
51	Exemption	SEZ Decree No.188. Art 40.3	For SEZ Developers: Construction of road, electricity system, water supply, wastewater treatment and waste disposal system shall be exempted from value added tax
52	Rate Relief	SEZ Decree No.188. Art 40.3	For SEZ Developers: Construction of any other infrastructure shall pay value added tax at fifty percent (50%) of the value added tax rate as specified in the Law on Value Added Tax.
53	Rate Relief	SEZ Decree No.188. Art 43.3	For SEZ Investors: the use of electricity and water in production: fifty percent (50%) of the value added tax rate specified in the Value Added Tax Law;

54	Rate Relief	SEZ Decree No.188. Art 43.3	For SEZ Investors: For the construction and development of infrastructures to support business operations of investors in sectors that are not 100% production for export, investors shall pay fifty percent (50%) of the value added tax rate as specified in the Value Added Tax Law.
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Table 5.2.2.1

- **VAT Tax Rate Summary Table**

Tax Unit	Tax Base	Tax Period	Tax Rate
Individual, legal entity or organization that operates a business under the VAT system	Defined in Article 11 of the VAT Law	Monthly basis	The benchmark VAT rate in Laos is currently 7% (entered into force from 01 January 2022)
			Export of goods and export of processed natural resources are charged VAT at 0%

Table 5.2.2.2

5.4.2.2 Define the Minimum Data table

As we analyzed at the beginning of this section, the minimum data on VAT Tax should include the following information:

No.	Column Name	Description	Calculation/Input/Remark
1	Uniq ID for all Record in system	ID Transactions in system	Mandatory , Unique Identify
2	TIN of Organization / person	Tax Identification Number of the enterprise.	Mandatory

3	Te VAT Provision (if have)	Provision of this Tax transaction	Option Only for double check
4	Date time appear Income (date time sale/buy...)	The time of transaction of goods/services that may generate tax.	Mandatory
5	Total VAT income (Invoice Amount LAK)	Total Amount of goods/services ... to be taxed	Mandatory
6	Tax rate (if have)	Tax rate	Option Only for double check
7	VAT TE value (if have for compare)	TE Value from other source...	Option Only for double check
8	Currency type/name, default is LAK	Currency	Option
9	Amount of VAT collected/PAID (if have)	The amount the taxpayer paid for this tax	Option to Calculate TE Value Remaining
10	TE VAT Tax Benchmark (if have)	TE Value benchmark from other source...	Option Only for double check
11	Variables to classification kind of income VAT tax	Variables are used to classify goods and thereby find the tax rate that will be applied. Please see the table below for details on the values of this variable.	Mandatory. Select a value from the list below to classify the goods/services.... that are subject to this tax.

5.4.2.3 Inherit data content from MS Excel files provided by MOF experts

We received a sample data file for reference from the MOF expert group, with the data structure described as follows:

Data structure in MS Excel file			Matching results (The results match what we need)	
No	Group	Column name	Compatible / or usable or not	Reason why NOT
1		TIN	YES	
2		Filing date	YES	
3		Description	NO	For reference only, cannot be used in calculation

				formulas
4	1 - Income and output VAT	Domestic VAT exempted (5)	NO	Only has statistical significance, comparison within the reference range in report and analytic. It is the result after tax calculate/ estimate benchmark. Cannot be used in calculation formulas / estimate benchmark
5		Abroad/SEZ VAT exempted (6)	NO	As above
6		Domestic 0% VAT (8)	NO	As above
7		Abroad/SEZ 0 % VAT (9)	NO	As above
8		Domestic 7% VAT (10)	NO	As above
9		Abroad/SEZ 7 % VAT (11)	NO	As above
10		Total VAT Collected (13)	NO	As above
11	2 - Expense and input VAT (Deductible VAT)	Exempted Domestic Expense (5)	NO	As above
12		Exempted Import Expense (6)	NO	As above
13		Domestic Expense 0% (8)	NO	As above
14		Import Expense 0% (9)	NO	As above
15		Domestic Expense 7% (10)	NO	As above
16		Import Expense 7% (11)	NO	As above
17		Total deductible VAT (15)	NO	As above
18	A - Output VAT (sales) transaction	Total sales VAT exemption (Domestic and abroad) (01)	NO	As above
19		Total exported sales and domestic sales at 0% VAT and abroad sales (03)	NO	As above
20		Total exported sales 7% VAT (07)	NO	As above
21		Total domestic sales 7% VAT (08)	NO	As above
22		Total (09)	NO	As above

(From MS Excel file named “Domestic VAT Template_import.xlsx”)

Quick review: There is some basic information but still missing important information to classify taxes by goods/services and find the corresponding tax rate. All data is just statistical results as output from sand box. No breakdown allowed to recalculate benchmark tax.

Note: The data highlighted in yellow is useful for VAT tax calculation and we will use it in the calculation formula.

And another sample data file with name “Domestic VAT Test.xlsx” with the data structure described as follows:

Data structure in MS Excel file		Matching results (The results match what we need)	
No	Column name	Compatible / or usable or not	Reason why NOT
1	Province	YES	
2	TIN	YES	
3	Name	NO	For reference only, cannot be used in calculation formulas
4	FilingType	NO	For reference only, cannot be used in calculation formulas
5	FilingPeriod	NO	For reference only, cannot be used in calculation formulas
6	InputDate	YES	
7	Description of the sale	NO	For reference only, cannot be used in calculation formulas
8	Rate	NO	For reference and double check/compare
9	Domestic sale exemption	NO	For reference and double check/compare
10	Domestic VAT TE	NO	For reference and double check/compare
11	# of Provision	NO	For reference and double check/compare

Quick review: There is more information (TIN,Datetime) available but still missing important information to classify taxes by goods/services and find the corresponding tax rate and Total VAT income/Invoice Amount. All data is just statistical results from the sandbox. No analysis is allowed to recalculate the standard tax. Provison data may allow for calculation support, but is still not enough to calculate the standard tax

And another sample data file with name “Customs, Excise, Import VAT Test.xlsx” with the data structure described as follows:

Province	TIN	No	BorderCode	Border	Type of Customs Declare	Process	F	Special Role	Import Document Number	Import Document Date	Assess Number	Assess Date	TryD Receipt No	TryD Receipt Date	Import/Export Name
Khammouan	580761167-9003335	R3C60		ດ່ານພາສີສາກົນນ່ວງ	IM	4000	480		10158	24/10/2022	10121	24/10/2022	221024000105	24/10/2022	ບ/ສ ຊີໄນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9001597	R1C10		ດ່ານພາສີສາກົນນ່ວງ	IM	4000	481		30686	17/09/2021	30658	18/09/2021	20210918490412	18/09/2021	ບ/ສ ຊີໄນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9005556	R1C10		ດ່ານພາສີສາກົນນ່ວງ	IM	4000	441		27812	26/06/2023	27559	26/06/2023	230626000412	26/06/2023	ບ/ສ ຊີໄນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9006979	R1C10		ດ່ານພາສີສາກົນນ່ວງ	IM	4000	000	ACN	32254	20/07/2023	31880	20/07/2023	230720000284	20/07/2023	ບ/ສ ຊີໄນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9004918	R1C10		ດ່ານພາສີສາກົນນ່ວງ	IM	4000	441		59925	28/11/2023	59642	29/11/2023	231129000065	29/11/2023	ບ/ສ ຊີໄນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9002132	R1C10		ດ່ານພາສີສາກົນນ່ວງ	IM	4000	481		22068	08/07/2021	21992	08/07/2021	202107081456624	08/07/2021	ບ/ສ ຊີໄນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ

Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	
Customs Declare TIN	Customs Declare Name	Export Country Code	Destination Country Code	Original Country Code	List	Goods HS	Goods Description	Quantity	Unit	Declare Weight	Actual Weight	Invoice Amount USD	Invoice Amount LAK	Customs	Ex
໑977704808900	ບສ ເອັກຊິມ ຈຳກັດຜູ້ດຽວໄຊເສດຖາ, ນ.ໄຊ	VN	LA	VN	1	3.924E+09	ຖ້ຳຜະລິດຕະພັນ(ຂະໜາດໂຕນ:90*90*125ລຸມ) ຈຳນວນ	12700	KGM	26774	26670	76200	1258671600	0	
໑625607046000	ບ/ສ ຮຸ່ງອາລຸນການຄ້າ ຂາອອກ-ຂາເຂົ້າບ	CN	LA	CN	1	3923909000	ຖົງຜະລິດຕະພັນ(ຂະໜາດໂຕນ) 415ໄປ້າ=8,300ອັນໃນ	15770	KGM	16060	15770	49800	477831000	0	
໑681630811000	ບໍລິສັດ ຂົນສົ່ງສາກົນ ຈຳກັດຈອມແກ້ວ, ນ	CN	LA	CN	1	8.426E+09	ເຄື່ອງໝັກໜັກໄຟຟ້າແບບຄານດຽວ LB5T-16.5Mພ້ອມ	3	NAR	31470	31134	45000	800775000	0	
໑681630811000	ບໍລິສັດ ຂົນສົ່ງສາກົນ ຈຳກັດຈອມແກ້ວ, ນ	CN	LA	CN	6	8.431E+09	ປອກຈັບດ້ານລຸ່ມ =12ອັນ=1໔໔ກັດ=68ກິໂລ	12	NAR	70	68	170	3247340	0	
໑681630811000		CN	LA	CN	2	7.216E+09	ເຫຼັກຮ່ອງ 160*63*6mm, 6M =72ອັນ=7.400ກິໂລ	7400	KGM	7400	7400	18500	384707500	0	
໑625607046000	ບ/ສ ຮຸ່ງອາລຸນການຄ້າ ຂາອອກ-ຂາເຂົ້າບ	CN	LA	CN	4	7208100000	ເຫຼັກແຜ່ນລາຍ = 1,077ກິໂລ	1077	KGM	1077	1077	862	8176932	0	

AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT
Excise	VAT	Profi t	RoadFun d	Total	AJ	Exemp_Custo ms	Exempt_Excise	Exempt_V AT	TE_Customs	TE_Excise	TE_VAT	Provision_Custo ms	Provision_Excise	Provision_VAT
0	0	0 0		0	Assessed	251734400	0	105728500	251734400		105728500	4000-480		4000-480
0	0	0 0		0	Assessed	95566200	0	57339800	95566200		57339800	4000-481		4000-481
0	0	0 0		0	Assessed	40038800	0	58857000	40038800		58857000	4000-441		4000-441
0	357300	252400 0		609700	Assessed	162400	0	0	162400			4000-000		
0	0	0 0		0	Assessed	19235400	0	28276100	19235400		28276100	4000-441		4000-441
0	0	0 0		0	Assessed	408900	0	858600	408900		858600	4000-481		4000-481

Data structure in MS Excel file		Matching results (The results match what we need)	
No	Column name	Compatible / or usable or not	Reason why NOT
1	Province	YES	
2	TIN	YES	
3	No	YES (use for ID_Transactions)	
4	BorderCode	NO	less relevant / redundant / used for other purposes
5	Border	NO	
6	TpyeofCustomsDeclare	NO	
7	Processtp	NO	
8	F	NO	
9	SpecialRole	NO	
10	ImportDocumentNumber	NO	less relevant / redundant / used for other purposes
11	ImportDocumentDate	YES (~ Input_date)	
12	AssessNumber	NO	
13	AssessDate	NO	
14	TryDReceiptNo	NO	
15	TryDReceiptDate	NO	
16	ImportExportName	NO	

17	CustomsDeclareTIN	NO	
18	CustomsDeclareName	NO	
19	ExportCountryCode	NO	
20	Destinationcountrycode	NO	
21	OriginalCountryCode	NO	
22	List	NO	
23	GoodsHS	NO	but use for HS CODE in Customs Tax)
24	GoodsDescription	NO	less relevant / redundant
25	Quantity	NO	
26	Unit	NO	
27	DeclareWeight	NO	
28	ActualWeight	NO	
29	InvoiceAmountUSD	NO	
30	InvoiceAmountLAK	YES (~ Total Income Value)	
31	Customs	NO	As a result for tax CUSTOMS, only use for reference/check in other scope
32	Excise	NO	As a result for tax EXCISE, only use for reference/check in other scope
33	VAT	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
34	Profit	NO	less relevant / redundant / only use for reference/check in other scope
35	RoadFund	NO	
36	Total	NO	
37	AJ	NO	
38	Exemp_Customs	NO	As a result for tax Customs, only use for reference/check in other scope
39	Exempt_Excise	NO	As a result for tax EXCISE, only use for reference/check in other scope
40	Exempt_VAT	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
41	TE_Customs		As a result for tax Customs, only use for reference/check in other scope
42	TE_Excise		As a result for tax EXCISE, only use for reference/check in other scope
43	TE_VAT	YES (~ use for	

		reference/DOUBLE check AFTER run benchmark)	
44	Provision_Customs	NO	We do not have a code table corresponding to "Provision_Customs"," Provision_Excise"," Provision_VAT", and have not been provided with a classification structure according to this code table.
45	Provision_Excise	NO	
46	Provision_VAT	NO	

Note: The data highlighted in yellow is useful for VAT tax calculation and we will use it in the calculation formula or double check result benchmark.

After careful consideration of each case and with an eye to future expansion, we have determined that the minimum data sheet (including both mandatory and optional information) should include at least the following information:

No.	Column Name	Description	Calculation/Input/Remark	The data provided by the experts is of similar meaning/usability.
1	Uniq ID for all Record in system	ID Transactions in system	Mandatory , Unique Identify	No
2	TIN of Organization / person	Tax Identification Number of the enterprise.	Mandatory	TIN
3	Te VAT Provision (if have)	Provision of this Tax transaction	Option Only for double check	
4	Date time appear Income (date time sale/buy...)	The time of transaction of goods/services that may generate tax.	Mandatory	ImportDocumentDate
5	Total VAT income (Invoice Amount LAK)	Total Amount of goods/services ... to be taxed	Mandatory	InvoiceAmountLAK or InvoiceAmountUSD
6	Tax rate (if have)	Tax rate	Option Only for double check	
7	VAT TE value (if have for compare)	TE Value from other source...	Option Only for double check	VAT
8	Currency type/name, default is LAK	Currency	Option	

9	Amount of VAT collected/PAID	The amount the taxpayer paid for this tax	Option to Calculate TE Value Remaining	
10	TE VAT Tax Benchmark (if have)	TE Value benchmark from other source...	Option Only for double check	TE_VAT
11	Variables to classification kind of income VAT tax	Variables are used to classify goods and thereby find the tax rate that will be applied. Please see the table below for details on the values of this variable.	Mandatory. Select a value from the list below to classify the goods/services.... that are subject to this tax.	Missing

Table 5.2.2.3

Conclusion: For VAT, the data provided by the expert (or import template) has a bit discrepancy in comparison to our system design. Expert data relies on the end result, like VAT Paid but less concerns about goods detail, we believe that the data of expert are based on external system that take care of such calculation (TaxRIS/ASYCUDA) completely. In the other hand our system is design so, that having information to classify taxable goods/services, we can calculate VAT Paid according to related provision. Nervelessness both method provide the same result, while the expert method is kind of shortcut and rely on the data from external system such as TaxRIS/ASYCUDA.

For our case, we will create a new variable (**Variable name / Type of the goods**) to fix this issue. Please see details below.

This is also the MS Excel file structure used in the VAT tax data import function.

Based on the "VAT Tax Rate Summary Table" and **Provision list provided by MOF expert (from MS Excel file)** above, which is provided by the expert, we have built specific variables, corresponding to the rules / provisions as follows:

Ids	TE Description	Variables to classification kind of income VAT tax	Tax Provision / TE #
1	Import of: all types of crop seeds, animals for breeding, animal sperms, vaccines, equipment and liquid nitrogen for storing vaccines and animal sperms, animal feeds, raw materials to produce animal feeds and vaccines;	VAT_Import_crop_seeds_animals_breeding_animal_sperms_vaccines	31

2	Import of: Raw materials used in the production of fertilizers, agroprocessing products, organic fertilizers, [scientific] fertilizers [and] pesticides that are not dangerous to the ecosystem, human and animal health and life	VAT_materials_fertilizers_agroprocessing_fertilizers_pesticides	32
3	Import of: Equipment and machinery used in agriculture;	VAT_Import_equipment_machinery_agriculture	33
4	Import of: Materials [and] equipment that could not be supplied or produced in Lao PDR and machinery that are used as fixed assets and used directly in production;	VAT_materials_equipment_not_supplied_lao_used_fixed_assets	34
5	Import of: Animal medicines, artificial organs for transplantation for animal bodies	VAT_Import_animal_medicines_artificial_organs_transplantation	35
6	Import of: Traditional medicines, artificial organs for transplantation for human bodies, human blood and supporting equipment for patients, the disabled and elderly;	VAT_Import_traditional_medicines_artificial_organs	36
7	Import of: Medical tools [and] equipment, [laboratory] equipments or wheelchairs for public services of hospitals, health care centers.	VAT_Import_medical_tools_equipment_wheelchairs_public_services	37
8	Import of: Vehicles serving professional activities, contributing to public benefits such as fire trucks, ambulances, vehicles equipped with repair facilities, outside television and radio broadcast vehicles and other professional vehicles.	VAT_Import_vehicles_serving_professional_activities_public	38
9	Domestic supply of: Unprocessed agricultural or preliminary processed products including peeled, grinded [and] milled;	VAT_Domestic_supply_unprocessed_agricultural_preliminary	39
10	Domestic supply of: All kinds of live or deceased animals, whole or in parts, that are unprocessed or preliminary processed to ensure freshness or non-perishability;	VAT_Domestic_supply_live_deceased_animals	40
11	Domestic supply of: afforestation, plantation of industrial, fruit and medicinal trees;	VAT_Domestic_supply_afforestation_plantation_fruit_medicinal	41
12	Domestic supply of: all types of crop seeds, animals for breeding, animal feeds, vaccines, raw materials to produce animal feeds and	VAT_Domestic_supply_crop_seeds_animals_breeding_feeds_vaccines	42

	vaccines;		
13	Domestic supply of: Raw materials used in the production of fertilizers, agroprocessing products, organic fertilizers, [scientific] fertilizers and pesticides that are not dangerous to the ecosystem, human and animal health and life;	VAT_Domestic_supply_raw_materials_fertilizers_agroprocessing	43
14	Domestic supply of: equipment and machinery used in agricultural activities;	VAT_Domestic_supply_equipment_machinery_agricultural	44
15	Domestic supply of: textbooks and learning materials	VAT_Domestic_supply_textbooks_learning_materials	45
16	The supply of education services by e.g. childcare centres, kindergartens, primary and secondary schools, vocational schools, colleges, academies, universities, sporting and athletic schools.	VAT_Supply_education_services	46
17	The examination, treatment, and diagnosis of humans and animals	VAT_examination_treatment_diagnosis_humans_animals	47
18	Animal medicines and artificial organs for transplantation of animal bodies	VAT_Animal_medicines_artificial_organs_transplantation	48
19	Traditional medicines, artificial organs for transplantation of human bodies, human blood, equipment or wheelchairs for patients, disabled and elderly;	VAT_medicines_artificial_organs_transplantation_human	49
20	Medical tools, equipment, [laboratory] analytical tools for hospitals, healthcare centers to serve society;	VAT_Medical_tools_equipment_hospitals_healthcare_society	50
21	For SEZ Developers: Construction of road, electricity system, water supply, wastewater treatment and waste disposal system shall be exempted from value added tax	VAT_SEZ_Developers_construction_road_electricity_water	51
22	For SEZ Developers: Construction of any other infrastructure shall pay value added tax at fifty percent (50%) of the value added tax rate as specified in the Law on Value Added Tax.	VAT_SEZ_Developers_For_Construction_any_other_infrastructure	52

23	For SEZ Investors: the use of electricity and water in production: fifty percent (50%) of the value added tax rate specified in the Value Added Tax Law;	VAT_SEZ_Investors_use_electricity_water_production	53
24	For SEZ Investors: For the construction and development of infrastructures to support business operations of investors in sectors that are not 100% production for export, investors shall pay fifty percent (50%) of the value added tax rate as specified in the Value Added Tax Law.	VAT_SEZ_Investors__For_construction_infrastructures_not_100	54
25	Others: Export of goods and export of processed natural resources are charged VAT at 0%	VAT_Export_goods_processed_natural_resources_0_percent	92

Table 5.2.2.4

In which: Tax Provision / TE: from No 31 to 54 based on prototype from MOF expert statistical file. And No 92 is for exception, it represents the "Other provision" of the VAT tax, it also has the same meaning as "Undefined" or "Unclassified".

In addition, there are some exceptions to VAT, for example:

- Before 2022, the standard default tax rate is 10%.
- From 01-01-2022 to 31-12-2022, the standard default tax rate is 7% according to the legal document "..."
- From 2024, the new default standard tax rate is 10% according to the legal document "...", although we have not been provided with official documentation on this matter.

To solve this problem, we have set up 03 more default variables with the meaning of representing the default tax rate value according to the change of the above documents. These variables are placed in the system parameter group and can change the tax rate value flexibly according to the decision of the Ministry of Finance by the system administrator.

System Variables name	Description	VAT rate default
Rate_Tax_VAT_standard_default_old	Rate of VATax by default before 2022	10%
Rate_Tax_VAT_standard_default_new_since_2022	Rate of VATax standard default based on New legal document (since 2022 to 2023)	7%
Rate_Tax_VAT_standard_default_new_since_2023	Rate of VATax standard default based on New legal document (since 2024)	10%

Table 5.2.2.5

This means that in case a tax has been recorded in the tax system but has not been classified under any tax group, the default tax rate will be applied over time according to the above system parameters.

For VAT, we apply the same classification method as profit tax. That is, using a pre-defined tax rule table with appropriate variables in the way that the experts of the Ministry of Finance have recommended for profit tax as described on **Table 5.2.2.4**

5.4.2.4 Create template file and build function on Dashboard of TaxETS system.

Below is some functional interface screens related to **VAT tax**

1. Import and manage VAT tax

From the perspective of system development, we plan to put all the sub-taxes in the VAT group into a single function. That means that this tax data can use the same data table and only differ in the specific data file or can be classified by adding a column representing the VAT sub-tax (e.g. domestic VAT, import VAT...). So far, we have not received complete information about all these sub-taxes, so we still respect the existing data in each sub-group and still keep the import functions on the database provided by the experts, as it will be useful for future reference and cross-checking (if any). And so, we build a new function to manage the actual VAT data without being bound by any sub-tax of VAT. This tax data management function will be a common template for all other sub-taxes (if any later). The new function is placed in the position as you see on the screen. Function name is VAT Tax (means VAT data actually currently)

Get Tax Data by Import from Excel								
Data Requirements to identify the type of Repository								
Data Requirements to estimate TE								
Profit Tax								
Excise Tax Actually								
Custom Tax Actually								
Individual Tax								
Salary Tax								
Domestic VAT								
For SEZ Developers								
For SEZ Investors								
Customs, Excise Tax,								
Import VAT from ASYCUDA								
Non-Tax: Land concession								
Non-Tax: Resource fee								
Non-Tax: Royalty fee								
VAT Tax								

VAT Actually								
Browse Data								
(Show 1 ~ 11 data from total 11 records)								
UNIQ ID FOR ALL RECORD IN SYSTEM %	TIN OF ORGANIZATION / PERSON %	TE VAT PROVISION %	DATE TIME APPEAR INCOME (DATE TIME SALE/BUY.) %	TOTAL VALUE INCOME %	TAX RATE (IF HAVE) %	VAT TE VALUE FOR COMPARE (IF HAVE) %	EXPORT OF GOODS AND EXPORT OF PROCESSED NATURAL RESOURCES ARE CHARGED VAT AT 0% %	VARIABLES TO CLASSIFICATION KIND OF INCOME TAX %
777	7567	5	30/11/-0001 00:00:00	3,423,423	9%		0	VAT_Animal_medicines_artificial_organ...
45	5555		02/09/2025 00:00:00	115,260,000	6%		0	VAT_Import_equipment_machinery_agriculture
5	4444		01/06/2021 00:00:00	1,871,120,832	10%		0	VAT_materials_equipment_not_supplied_lao_us...
4	3333		01/05/2021 00:00:00	10,008,773,452	7%		0	VAT_Import_equipment_machinery_agriculture
2	2222		01/01/2020 00:00:00	5,000,000	7%		0	VAT_Import_equipment_machinery_agriculture

MS Excel template file as described in **Table 5.2.2.3** Define the Minimum Data table. Users can download from the Import function interface.

2. Run Benchmark VAT Tax

TaxETS

- System
- Data Dictionary
- Get Tax Data by Import from Excel
- Get Tax Data by API
- Run Tax Expenditure
 - Run Profit Tax Expenditure
 - Run Individual Tax Expenditure
 - Run Excise Tax Expenditure
 - Run VAT Tax Expenditure**
 - Run Custom Tax Expenditure

Run VAT Tax Expenditure

TIN: Start Date: End Date:

[View data](#)

[Search](#) [Run](#)

	UNIQ ID FOR ALL RECORD IN SYSTEM	TIN	INPUT DATE	TOTAL VAT INCOME	VAT TAX RATE	TE VALUE	CURRENCY
☐	1233	222	01/01/2021 07:00:00	400,000			LAK
☐	228801	7123456789-5858	01/01/2020 07:00:00	5,000,000			LAK
☐	1234	333	01/05/2021 07:00:00	10,008,773,452			LAK
☐	1235	580761167-900	01/06/2021 07:00:00	1,871,120,832			LAK
☐	1236	583429006-900	02/09/2025 07:00:00	115,260,000			LAK
☐	1	1111	01/01/2021 07:00:00	400,000	0%		LAK
☐	2	2222	01/01/2020 07:00:00	5,000,000	7%		LAK

Input:

- TIN
- Start date and End date: Date time when you want to start running Benchmark tax calculations for the selected TIN (that means from time - to time)

Output: The system will automatically search for entered tax data that meets the above conditions and calculate the standard tax value (if met).

2. View Result Benchmark VAT Tax

Run VAT Tax Expenditure Result Data										Dashboard » Run VAT Tax Expenditure Result Data
Browse Data										Search...
(Show 1 ~ 1 data from total 1 records)										
UNIQ ID FOR ALL RECORD IN SYSTEM ↑↓	TIN OF ORGANIZATION / PERSON ↑↓	TE VAT PROVISION ↑↓	DATE TIME APPEAR ↑↓	TOTAL VALUE INCOME ↑↓	TAX RATE ↑↓	VAT TE ↑↓	DATETIME CREATED ↑↓	USER CREATE/UPDATE ↑↓	NOTE ↑↓	ACTION
1234	333	52	01/05/2021 00:00:00	10,008,773,452	5.00%	500,438,673	15/11/2025 21:24:30			
										Items per page: 10 ▼

Output:

- + Tax rate
- + VAT TE value for compare (VAT benchmark value)
- + Amount of VAT tax collected/PAID (if have)
- + TE Value (VAT TE = VAT Benchmark - VAT PAID)
- + Datetime RUN
- + User ID RUN

5.4.2.5 VAT Test Result.

Here is an example from TaxETS system:

Run VAT Tax Expenditure Result Data								Dashboard » Run VAT Tax Expe
Browse Data								Search 5807611
(Show 1 ~ 1 data from total 1 records)								
UNIQ ID FOR ALL RECORD IN SYSTEM ↑↓	TIN OF ORGANIZATION / PERSON ↑↓	TE VAT PROVISION ↑↓	DATE TIME APPEAR ↑↓	TOTAL VALUE INCOME ↑↓	TAX RATE ↑↓	VAT TE ↑↓	DATETIME CREATED ↑↓	
1235	580761167-900	92	01/06/2021 00:00:00	1,871,120,832	10.00%	187,112,083	19/11/2025 11:36:40	

And here is data provided by Expert:

Province	TIN	FilingPeriod	InputDate	Description of the sale	Rate	Domestic sale exemption	Domestic VAT TE
Department	880782489-900	10/1	09/11/2020		10	1,617,992,605	161,799,260.50
Department	580761167-900	06/2021	16/07/2021		10	1871120832	187,112,083.20
Department	543163802-900	05/2021	14/06/2021		10	10008773452	1,000,877,345.20

So, the figure of 187,112,083,20 is exactly the same.

5.4.3 Excise Tax

5.4.3.1 Excise calculation rule prototype and legal documents

- **Legal documents:**

- + *Investment Law 2016_English version.pdf*
- + *List of Investment Promotion Activity (English version)*
- + *Presidential Edict 003 (2023) New Excise Tax.pdf*

- **Provision list provided by MOF expert (from MS Excel file)**

TE #	Type	Legal Basis	TE Description
57	Exemption	Excise Tax Law Art 12(5)	Rescue vehicles, rescue airplanes, rescue boats, ambulances, fire trucks, and specialized service vehicles for the National Defence Forces and security forces;
58	Exemption	Excise Tax Law Art 12(6)	Machinery Used for agricultural production
59	Exemption	Excise Tax Law Art 12(7)	Heavy machinery such as backhoe loader, drum rollers, excavators, motor graders, bulldozer, A truck equipped for pile foundation work that directly contribute to production and are considered as fixed assets
60	Exemption	Excise Tax Law Art 12(10)	Motorcycles using clean energy

61	Exemption	Excise Tax Law Art 12(11)	Vehicle components imported for assembly in domestic factory system
62	Rate Relief	Excise Tax Law Art 15(2.1.1.2)	Passenger vehicles using hybrid energy pay 50% of the applicable rates based on engine size for fuel energy (outlined in Art. 12 (2.1.1.1)).
63	Rate Relief	Excise Tax Law Art 15(2.1.1.3)	Passenger transportation vehicles using clean energy pay a rate of 3%.
64	Rate Relief	Excise Tax Law Art 15(2.1.2)	Cargo transportation vehicles like: normal 2-door pickup trucks and semi-automatic 2-door pickup trucks using clean energy pay a rate of 3%
65	Rate Relief	Excise Tax Law Art 15(2.1.3)	Other small passenger and cargo transportation vehicles using clean energy pay a rate of 3%
66	Rate Relief	Excise Tax Law Art 15(2.2)	Medium passenger and cargo transportation vehicles using clean energy pay a rate of 3%
67	Rate Relief	Excise Tax Law Art 15 (2.3)	Large passenger and cargo transportation vehicles using clean energy pay a rate of 3%
68	Rate Relief	Excise Tax Law Art. 15 (last paragraph)	Vehicles utilizing fuel energy, manufactured or assembled within the domestic factory system in CKD form are subject to a 5% excise tax rate
69	Rate Relief	Excise Tax Law Art. 15 (last paragraph)	Vehicles utilizing fuel energy, manufactured or assembled within the domestic factory system in IKD form are subject to a 3% excise tax rate

Table 5.2.3.1

- Excise Tax Rate Summary Table

Tax Unit	Tax Base	Tax Period	Tax Rate
(i) goods importer, (ii) manufacturers of goods, (iii) service providers or (iv) buyers of services from non-resident households.	(i) For imported goods, the cost, insurance and freight (CIF) value, plus any applicable customs duties. (ii) For locally supplied goods and services, the value at the time of supply.	Importer: at the time of customs clearance. Domestic manufacturers and service providers: monthly basis and not later than 15th day of the following month.	There is no single benchmark tax rate for excise tax. The benchmark rates for each product category and sub-category are as listed in the Excise Tax Law Article 15 and Presidential Edict No. 003/P of October 2023

Table 5.2.3.2

5.4.3.2 Define the Minimum Data table

As we analyzed at the beginning of this section, the minimum data on Excise Tax should include the following information:

No.	Column Name	Description	Calculation/Input/Remark
1	Uniq ID for all Record in system	ID Transactions in system	Mandatory , Unique Identify
2	TIN of Organization / person	Tax Identification Number of the enterprise.	Mandatory
3	TE Excise provision (if have)	Provision of this Tax transaction	Option Only for double check
4	Date time appear Income (date time sale/buy..)	The time of transaction of goods/services that may generate tax.	Mandatory

5	Total value income	Total Amount of goods/services ... to be taxed	Mandatory
6	Tax rate (if have)	Tax rate	Option Only for double check
7	Excise TE value for compare (if have)	TE Value from other source...	Option Only for double check
8	Currency type/name, default is LAK	Currency	Option
9	Amount of Excise collected/PAID	The amount the taxpayer paid for this tax	Option to Calculate TE Value Remaining
10	Excise Tax estimate by Rules	TE Value benchmark from other source...	Option Only for double check
11	Variables to classification of Excise tax (Type of the goods)	Variables are used to classify goods and thereby find the tax rate that will be applied. Please see the table below for details on the values of this variable.	Mandatory. Select a value from the list below to classify the goods/services.... that are subject to this tax.

Table 5.2.3.3

5.4.3.3 Inherit data content from MS Excel files provided by MOF experts

We received a sample data file for reference from the MOF expert group. Data file with name “Customs, Excise, Import VAT Test.xlsx” with the data structure described as follows:

Province	TIN	No	BorderCode	Border	Type of Customs Declaration	Process	F	Special Role	Import Document Number	Import Document Date	Assess Number	Assess Date	TryDReceiptNo	TryDReceiptDate	ImportExportName
Khammouan	580761167-9003335	R3C60	ດ່ານພາສີສາກົນນໍ້າ	IM	4000	480			10158	24/10/2022	10121	24/10/2022	221024000105	24/10/2022	ບ/ສ ຊີໂນ-ອາກຣີໂບຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9001597	R1C10	ດ່ານພາສີສາກົນນໍ້າ	IM	4000	481			30686	17/09/2021	30658	18/09/2021	20210918490412	18/09/2021	ບ/ສ ຊີໂນ-ອາກຣີໂບຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9005556	R1C10	ດ່ານພາສີສາກົນນໍ້າ	IM	4000	441			27812	26/06/2023	27559	26/06/2023	230626000412	26/06/2023	ບ/ສ ຊີໂນ-ອາກຣີໂບຕາສ ຈຳກັດນາໄຊໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9006979	R1C10	ດ່ານພາສີສາກົນນໍ້າ	IM	4000	000	ACN		32254	20/07/2023	31880	20/07/2023	230720000284	20/07/2023	ບ/ສ ຊີໂນ-ອາກຣີໂບຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9004918	R1C10	ດ່ານພາສີສາກົນນໍ້າ	IM	4000	441			59925	28/11/2023	59642	29/11/2023	231129000065	29/11/2023	ບ/ສ ຊີໂນ-ອາກຣີໂບຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-9002132	R1C10	ດ່ານພາສີສາກົນນໍ້າ	IM	4000	481			22068	08/07/2021	21992	08/07/2021	202107081456624	08/07/2021	ບ/ສ ຊີໂນ-ອາກຣີໂບຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ

Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF
CustomsDeclareTIN	CustomsDeclareName	ExportCountryCode	Destinationcountrycode	OriginalCountryCode	List	GoodsHS	GoodsDescription	Quantity	Unit	Declare Weight	ActualWeight	InvoiceAmountUSD	InvoiceAmountLAK	Customs	Excise
977704808900	ບສ ເອັກຊິມ ຈຳກັດຜູ້ດຽວໄຊເສດຖາ, ນຳ	VN	LA	VN	1	3.924E+09	ຖ້ຳຜະລິດຕະພັນ(ຂະໜາດໂຕນ:90*90*125ລຸມ) ຈຳນວນ	12700	KGM	26774	26670	76200	1258671600	0	
625607046000	ບ/ສ ຮຸ່ງອາລຸນການຄ້າ ຂາອອກ-ຂາເຂົ້າບ	CN	LA	CN	1	3923909000	ຖ້ຳຜະລິດຕະພັນ(ຂະໜາດໂຕນ) 415໒ປີາ=8,300ອັນໃນ	15770	KGM	16060	15770	49800	477831000	0	
681630811000	ບໍລິສັດ ຂົນສົ່ງສາກົນ ຈຳກັດຈອມແກ້ວ, ນ	CN	LA	CN	1	8.426E+09	ເຄື່ອງໝັກໜັກໄຟຟ້າແບບຄານດຽວ LB5T-16.5Mພ້ອມ	3	NAR	31470	31134	45000	800775000	0	
681630811000	ບໍລິສັດ ຂົນສົ່ງສາກົນ ຈຳກັດຈອມແກ້ວ, ນ	CN	LA	CN	6	8.431E+09	ປອກຈັບດ້ານລຸ່ມ =12ອັນ=1ແກ້ວ=68ກິໂລ	12	NAR	70	68	170	3247340	0	
681630811000		CN	LA	CN	2	7.216E+09	ເຫລັກຮ່ອງ 160*63*6mm,6M =72ອັນ=7.400ກິໂລ	7400	KGM	7400	7400	18500	384707500	0	
625607046000	ບ/ສ ຮຸ່ງອາລຸນການຄ້າ ຂາອອກ-ຂາເຂົ້າບ	CN	LA	CN	4	7208100000	ເຫຼັກແຜ່ນລາຍ = 1,077ກິໂລ	1077	KGM	1077	1077	862	8176932	0	

AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT
Excise	VAT	Profi t	RoadFun d	Total	AJ	Exempt_Custo ms	Exempt_Excise	Exempt_VAT	TE_Customs	TE_Excise	TE_VAT	Provision_Custo ms	Provision_Excise	Provision_VAT
0	0	0 0		0	Assessed	251734400	0	105728500	251734400		105728500	4000-480		4000-480
0	0	0 0		0	Assessed	95566200	0	57339800	95566200		57339800	4000-481		4000-481
0	0	0 0		0	Assessed	40038800	0	58857000	40038800		58857000	4000-441		4000-441
0	357300	252400 0		609700	Assessed	162400	0	0	162400			4000-000		
0	0	0 0		0	Assessed	19235400	0	28276100	19235400		28276100	4000-441		4000-441
0	0	0 0		0	Assessed	408900	0	858600	408900		858600	4000-481		4000-481

Data structure in MS Excel file		Matching results (The results match what we need)	
No	Column name	Compatible / or usable or not	Reason why NOT
1	Province	YES	
2	TIN	YES	
3	No	YES (use for ID_Transactions)	

4	BorderCode	NO	less relevant / redundant / used for other purposes
5	Border	NO	
6	TpyeofCustomsDeclare	NO	
7	Processtp	NO	
8	F	NO	
9	SpecialRole	NO	
10	ImportDocumentNumber	NO	
11	ImportDocumentDate	YES (~ Input_date)	
12	AssessNumber	NO	less relevant / redundant / used for other purposes
13	AssessDate	NO	
14	TryDReceiptNo	NO	
15	TryDReceiptDate	NO	
16	ImportExportName	NO	
17	CustomsDeclareTIN	NO	
18	CustomsDeclareName	NO	
19	ExportCountryCode	NO	
20	Destinationcountrycode	NO	
21	OriginalCountryCode	NO	
22	List	NO	but use for HS CODE in Customs Tax)
23	GoodsHS	NO	
24	GoodsDescription	NO	less relevant / redundant
25	Quantity	NO	
26	Unit	NO	
27	DeclareWeight	NO	
28	ActualWeight	NO	
29	InvoiceAmountUSD	NO	
30	InvoiceAmountLAK	YES (~ Total Income Value)	
31	Customs	NO	As a result for tax CUSTOMS, only use for reference/check in other scope
32	Excise	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
33	VAT		As a result for tax VAT, only use for reference/check in other scope
34	Profit	NO	less relevant / redundant / only use for reference/check in other scope
35	RoadFund	NO	
36	Total	NO	

37	AJ	NO	
38	Exemp_Customs	NO	As a result for tax Customs, only use for reference/check in other scope
39	Exempt_Excise	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
40	Exempt_VAT	NO	As a result for tax VAT, only use for reference/check in other scope
41	TE_Customs	NO	As a result for tax Customs, only use for reference/check in other scope
42	TE_Excise	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
43	TE_VAT	NO	As a result for tax VAT, only use for reference/check in other scope
44	Provision_Customs	NO	We do not have a code table corresponding to "Provision_Customs", "Provision_Excise", "Provision_VAT", and have not been provided with a classification structure according to this code table.
45	Provision_Excise	NO	
46	Provision_VAT	NO	

Table 5.2.3.4

Note: The data highlighted in yellow is useful for VAT tax calculation and we will use it in the calculation formula or double check result benchmark.

After careful consideration of each case and with an eye to future expansion, we have determined that the minimum data sheet (including both mandatory and optional information) should include at least the following information:

No.	Column Name	Description	Calculation/Input/Remark	The data provided by the experts is of similar meaning/usability.
1	Uniq ID for all Record in system	ID Transactions in system	Mandatory , Unique Identify	No
2	TIN of Organization / person	Tax Identification Number of the enterprise.	Mandatory	TIN
3	TE Excise provision (if have)	Provision of this Tax transaction	Option Only for double check	
4	Date time appear Income (date time sale/buy..)	The time of transaction of goods/services that may generate tax.	Mandatory	ImportDocumentDate
5	Total value income	Total Amount of goods/services ... to be taxed	Mandatory	InvoiceAmountLAK or InvoiceAmountUSD
6	Tax rate (if have)	Tax rate	Option Only for double check	
7	Excise TE value for compare (if have)	TE Value from other source...	Option Only for double check	Exempt_Excise
8	Currency type/name, default is LAK	Currency	Option	
9	Amount of Excise collected/PAID	The amount the taxpayer paid for this tax	Option to Calculate TE Value Remaining	
10	Excise Tax estimate by Rules	TE Value benchmark from other source...	Option Only for double check	TE_Excise
11	Variables to classify Excise tax (Type of the goods)	Variables are used to classify goods and thereby find the tax rate that will	Mandatory. Select a value from the list below to classify the goods/services.... that are	Missing

		be applied. Please see the table below for details on the values of this variable.	subject to this tax.	
--	--	--	----------------------	--

Table 5.2.3.5

Conclusion: For Excise tax, the data provided by the expert (or import template) has a bit discrepancy in comparison to our system design. Expert data relies on the end result, like Excise Tax Paid but less concerns about the detail of goods detail, we believe that the data is based on ASYCUDA system, that has made this calculation already. In the other hand our system is design so, that having information to classify taxable goods/services, we can calculate Excise Tax Paid according to related provision. Nervelessness both method provide the same result, while the expert method is kind of shortcut and rely on the data from external system such as ASYCUDA.

For our system, we will create a new variable (**Variable name / Type of the goods**) to fix this issue. Please see details below.

This is also the MS Excel file structure used in the Excise tax data import function.

Based on the "Excise Tax Rate Summary Table" and Provision list provided by MOF expert (from MS Excel file) above, which is provided by the expert, we have built specific variables, corresponding to the rules / provisions as follows:

ID Type	Type of goods	Variables to classify Excise tax (Type of the goods)	Tax rate	Effective date from	Effective date to	Tax Provision
1	Gasoline (super)	Gasoline_super	35	Unlimit	Unlimit	93
2	Gasoline (normal)	Gasoline_normal	30	Unlimit	Unlimit	93
3	Diesel	Diesel	20	Unlimit	Unlimit	93
4	Jet fuel	Jet_fuel	8	Unlimit	Unlimit	93
5	Lubricants, hydraulic oil, grease, and brake oil	Lubricants_hydraulic_oil_grease_and_bra ke_oil	5	Unlimit	Unlimit	93
6	Motorcycles	Motorcycles	5	Unlimit	Unlimit	93
7	•Passenger and transport vehicles	Passenger_and_transport_vehicles	10	Unlimit	Unlimit	93
8	• Sound systems	Sound_systems	20	Unlimit	Unlimit	93

9	• Decorative accessories	Decorative_accessories	15	Unlimit	Unlimit	93
10	• Alcohol or alcoholic beverages over 23 degrees	Alcohol_beverages_over_23_degrees	70	Unlimit	09-OCT-2023	93
11	• Alcohol or alcoholic beverages with 23 degrees or below	Alcohol_beverages_23_degrees_or_below	60	Unlimit	09-OCT-2023	93
12	6. Beer:	Beer	50	Unlimit	09-OCT-2023	93
13	• Cigars, packed or cased tobacco and other types of tobacco	Cigars_packed_cased_tobacco_other_tobacco	50	Unlimit	09-OCT-2023	93
14	• Electric cigarettes, nicotine compounds for use with electric cigarettes	Electric_cigarettes_nicotine_compounds	60	Unlimit	09-OCT-2023	93
15	• Shredded tobacco	Shredded_tobacco	35	Unlimit	09-OCT-2023	93
16	8. Gas for vehicles	Gas_for_vehicles	10	Unlimit	Unlimit	93
17	• Soft drinks, sodas, coffee drinks, and other similar drinks	Soft_drinks_sodas_coffee_drinks_and_other_similar_drinks	5	Unlimit	Unlimit	93
18	• Stimulant drinks	Stimulant_drinks	10	Unlimit	Unlimit	93
19	10. Crystal items or crystal adornments	Crystal_items_or_crystal_adornments	20	Unlimit	Unlimit	93
20	11. Carpets made from wool and animal parts	Carpets_made_from_wool_and_animal_parts	15	Unlimit	Unlimit	93
21	12. Perfume, accessories for beauty and beauty supplements	Perfume_accessories_for_beauty_and_beauty_supplements	20	Unlimit	Unlimit	93
22	13. Playing cards and gambling materials if authorized by the relevant sectors	Playing_cards_and_gambling_materials_if_authorized_by_the_relevant_sectors	90	Unlimit	Unlimit	93
23	14. Traditional rockets, fireworks, firecrackers if authorized by the relevant sectors	Traditional_rockets_fireworks_firecrackers_if_authorized_by_the_relevant_sectors	80	Unlimit	Unlimit	93
24	15. Speedboats, yachts, motorboats	Speedboats_yachts_motorboats	20	Unlimit	Unlimit	93

25	16. Satellite television signal receivers	Satellite_television_signal_receivers	15	Unlimit	Unlimit	93
26	17. Audio-video players, cameras, telephones, audio-video recorders, musical instruments	Audio_video_players_cameras_telephones_audio_video_recorders_musical_instruments	10	Unlimit	Unlimit	93
27	18. Aerial drones, aerial cameras, parachutes, and small jet engines	Aerial_drones_aerial_cameras_parachutes_and_small_jet_engines	20	Unlimit	Unlimit	93
28	19. Billiard tables, snooker tables, bowling equipment, football playing tables	Billiard_tables_snooker_tables_bowling_equipment_football_playing_tables	30	Unlimit	Unlimit	93
29	20. Coin game machine and all kinds of game players	Coin_game_machine_and_all_kinds_of_game_players	35	Unlimit	Unlimit	93
30	1. Entertainment: Dancing clubs, discotheques karaoke	Entertainment_Dancing_clubs_discotheques_karaoke	35	Unlimit	Unlimit	93
31	2. Bowling activities	Bowling_activities	20	Unlimit	Unlimit	93
32	3. Beauty surgery activities	Beauty_surgery_activities	10	Unlimit	Unlimit	93
33	4. Phone services, fiber optic television, digital television signals	Phone_services_fiber_optic_television_digital_television_signals	5	Unlimit	Unlimit	93
34	5. Internet services	Internet_services	3	Unlimit	Unlimit	93
35	6. Golfing services	Golfing_services	20	Unlimit	Unlimit	93
36	7. Lottery services	Lottery_services	25	Unlimit	Unlimit	93
37	8. Casino services, poker machines, and all kinds of game machines	Casino_services_poker_machines_and_all_kinds_of_game_machines	50	Unlimit	Unlimit	93
38	9. Motor racing, horse racing, and cockfighting	Motor_racing_horse_racing_and_cockfighting	25	Unlimit	Unlimit	93
39	10. Ballooning, parachuting and paragliding, and aerial photography services	Ballooning_parachuting_and_paragliding_and_aerial_photography_services	10	Unlimit	Unlimit	93
40	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_110_cc_and_lower_CBU	10	Unlimit	Unlimit	93

41	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_110_cc_and_lower_CKD	6	Unlimit	Unlimit	93
42	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_110_cc_and_lower_IDK	3	Unlimit	Unlimit	93
43	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_111_150_cc_CBU	20	Unlimit	Unlimit	93
44	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_111_150_cc_CKD	20	Unlimit	Unlimit	93
45	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_151_200_cc_CBU	25	Unlimit	Unlimit	93
46	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_151_200_cc_CKD	25	Unlimit	Unlimit	93
47	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_200_250_cc	35	Unlimit	Unlimit	93
49	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_250_500_cc_CBU	70	Unlimit	Unlimit	93
50	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_250_500_cc_CKD	70	Unlimit	Unlimit	93
51	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_500_800_cc_CBU	90	Unlimit	Unlimit	93
52	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_500_800_cc_CKD	90	Unlimit	Unlimit	93
53	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_800_cc_or_more_CBU	100	Unlimit	Unlimit	93
54	1.1.1 Gas- or diesel-powered motorcycles	With_engine_volume_of_800_cc_or_more_CKD	100	Unlimit	Unlimit	93
55	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1000_cc_and_lower_CBU	25	Unlimit	Unlimit	93

56	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1000_cc_and_lower_CKD	25	Unlimit	Unlimit	93
57	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1000_cc_and_lower_IKD	25	Unlimit	Unlimit	93
58	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1000_1600_cc_CBU	30	Unlimit	Unlimit	93
59	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1000_1600_cc_CKD	30	Unlimit	Unlimit	93
60	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1000_1600_cc_IKD	30	Unlimit	Unlimit	93
61	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1600_2000_cc_CBU	35	Unlimit	Unlimit	93
62	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1600_2000_cc_CKD	35	Unlimit	Unlimit	93
63	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_1600_2000_cc_IKD	35	Unlimit	Unlimit	93

64	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_2000_2500_cc_CBU	40	Unlimit	Unlimit	93
65	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_2000_2500_cc_CKD	40	Unlimit	Unlimit	93
66	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_2000_2500_cc_IKD	40	Unlimit	Unlimit	93
67	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_2500_3000_cc_CBU	45	Unlimit	Unlimit	93
68	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_2500_3000_cc_CKD	45	Unlimit	Unlimit	93
69	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_2500_3000_cc_IKD	45	Unlimit	Unlimit	93
70	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_3000_4000_cc_CBU	70	Unlimit	Unlimit	93
71	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_3000_4000_cc_CKD	70	Unlimit	Unlimit	93
72	2.1.1 Transportation vehicles like:	With_engine_volume_of_3000_4000_cc_	70	Unlimit	Unlimit	93

	cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	IKD				
73	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_4000_5000_cc_CBU	85	Unlimit	Unlimit	93
74	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_4000_5000_cc_CKD	85	Unlimit	Unlimit	93
75	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_4000_5000_cc_IDK	85	Unlimit	Unlimit	93
76	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_5000_cc_or_more_CBU	90	Unlimit	Unlimit	93
77	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_5000_cc_or_more_CKD	90	Unlimit	Unlimit	93
78	2.1.1 Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered	With_engine_volume_of_5000_cc_or_more_IKD	90	Unlimit	Unlimit	93
79	2.1.1.2 Hybrid passenger vehicles	Hybrid_passenger_vehicles_CBU	12.5	Unlimit	Unlimit	93
80	2.1.1.2 Hybrid passenger vehicles	Hybrid_passenger_vehicles_CKD	12.5	Unlimit	Unlimit	93
81	2.1.1.2 Hybrid passenger vehicles	Hybrid_passenger_vehicles_IKD	12.5	Unlimit	Unlimit	93

82	2.1.1.3 Electric passenger vehicles	Electric_passenger_vehicles_CBU	3	Unlimit	Unlimit	93
83	2.1.1.3 Electric passenger vehicles	Electric_passenger_vehicles_CKD	3	Unlimit	Unlimit	93
84	2.1.2 Two-door and two-and-a-half door pickups:	Two_door_Gas_or_diesel_powered_CBU	15	Unlimit	Unlimit	93
85	2.1.2 Two-door and two-and-a-half door pickups:	Two_door_Gas_or_diesel_powered_CKD	15	Unlimit	Unlimit	93
86	2.1.2 Two-door and two-and-a-half door pickups:	Two_door_Electric_powered_CBU	3	Unlimit	Unlimit	93
87	2.1.2 Two-door and two-and-a-half door pickups:	Two_door_Electric_powered_CKD	3	Unlimit	Unlimit	93
88	2.1.3 Small passenger and transport vehicles	Small_passenger_Gas_or_diesel_powered_CBU	10	Unlimit	Unlimit	93
89	2.1.3 Small passenger and transport vehicles	Small_passenger_Gas_or_diesel_powered_CKD	10	Unlimit	Unlimit	93
90	2.1.3 Small passenger and transport vehicles	Small_passenger_Electric_powered_CBU	3	Unlimit	Unlimit	93
91	2.1.3 Small passenger and transport vehicles	Small_passenger_Electric_powered_CKD	3	Unlimit	Unlimit	93
92	2.2 Medium passenger and transport vehicles	Medium_passenger_Gas_or_diesel_powered_CBU	8	Unlimit	Unlimit	93
93	2.2 Medium passenger and transport vehicles	Medium_passenger_Gas_or_dieselpowered_CKD	8	Unlimit	Unlimit	93
94	2.2 Medium passenger and	Medium_passenger_Electric_powered_C	3	Unlimit	Unlimit	93

	transport vehicles	BU				
95	2.2 Medium passenger and transport vehicles	Medium_passenger_Electric_powered_CKD	3	Unlimit	Unlimit	93
96	2.3 Large passenger and transport vehicles	Large_passenger_Gas_or_diesel_powered_CBU	5	Unlimit	Unlimit	93
97	2.3 Large passenger and transport vehicles	Large_passenger_Gas_or_dieselpowered_CKD	5	Unlimit	Unlimit	93
98	2.3 Large passenger and transport vehicles	Large_passenger_Electric_powered_CBU	3	Unlimit	Unlimit	93
99	2.3 Large passenger and transport vehicles	Largepassenger_Electric_powered_CKD	3	Unlimit	Unlimit	93
100	Cars that use fuel, manufactured or assembled in domestic factories in CKD form.	Cars_that_use_fuel_manufactured_or_assembled_in_domestic_factories_in_CKD_form.	5	Unlimit	Unlimit	93
101	Cars that use fuel, manufactured or assembled in domestic factories in IKD form.	Cars_that_use_fuel_manufactured_or_assembled_in_domestic_factories_in_IKD_form.	3	Unlimit	Unlimit	93
102	2.4	Refrigerated_vans_trucks_transporting_liquids_gas_cement_powder_or_fuel_water_vehicles_cranes_forklift_trucks_cement_mixers_waste_suction_vehicles_all_kinds_of_spraying_vehicles_and_camper_vans_CKD	10	Unlimit	Unlimit	93
103	2.4	Refrigerated_vans_trucks_transporting_liquids_gas_cement_powder_or_fuel_water_vehicles_cranes_forklift_trucks_cement_mixers_waste_suction_vehicles_all_kinds_of_spraying_vehicles_and_camper_vans_IDK	10	Unlimit	Unlimit	93
104	2.5	articulated_trucks_and_semiarticulated_trucks_CBU	5	Unlimit	Unlimit	93

105	2.5	articulated_trucks_and_semiarticulated_trucks_CKD	5	Unlimit	Unlimit	93
106	2.6	Golf_carts_and_vehicles_used_on_golf_courses_CBU	10	Unlimit	Unlimit	93
108	2.7	Sightseeing_passenger_cars_CBU	5	Unlimit	Unlimit	93
109	2.7	Sightseeing_passenger_cars_CKD	5	Unlimit	Unlimit	93
110	2.8	ATVs_and_Gokarts_CBU	25	Unlimit	Unlimit	93
111	2.8	ATVs_and_Gokarts_CKD	25	Unlimit	Unlimit	93
112	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1000_cc_and_lower_CBU_New_Presidential_Edict	31	09-OCT-2023	Unlimit	93
113	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1000_cc_and_lower_CKD_New_Presidential_Edict	31	09-OCT-2023	Unlimit	93
114	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1000_cc_and_lower_IKD_New_Presidential_Edict	31	09-OCT-2023	Unlimit	93
115	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-	With_engine_volume_of_1000_1600_cc_CBU_New_Presidential_Edict	41	09-OCT-2023	Unlimit	93

	powered (Presidential Edict 003 (2023) New Excise Tax)					
116	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1000_1600_cc_CKD_New_Presidential_Edict	41	09-OCT-2023	Unlimit	93
117	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1000_1600_cc_IKD_New_Presidential_Edict	41	09-OCT-2023	Unlimit	93
118	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1600_2000_cc_CBU_New_Presidential_Edict	56	09-OCT-2023	Unlimit	93
119	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1600_2000_cc_CKD_New_Presidential_Edict	56	09-OCT-2023	Unlimit	93
120	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_1600_2000_cc_IKD_New_Presidential_Edict	56	09-OCT-2023	Unlimit	93
121	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_2000_2500_cc_CBU_New_Presidential_Edict	66	09-OCT-2023	Unlimit	93

122	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_2000_2500_cc_CKD_New_Presidential_Edict	66	09-OCT-2023	Unlimit	93
123	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_2000_2500_cc_IKD_New_Presidential_Edict	66	09-OCT-2023	Unlimit	93
124	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_2500_3000_cc_CBU_New_Presidential_Edict	82	09-OCT-2023	Unlimit	93
125	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_2500_3000_cc_CKD_New_Presidential_Edict	82	09-OCT-2023	Unlimit	93
126	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_2500_3000_cc_IKD_New_Presidential_Edict	82	09-OCT-2023	Unlimit	93
127	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_3000_4000_cc_CBU_New_Presidential_Edict	127	09-OCT-2023	Unlimit	93

128	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_3000_4000_cc_CKD_New_Presidential_Edict	127	09-OCT-2023	Unlimit	93
129	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_3000_4000_cc_IKD_New_Presidential_Edict	127	09-OCT-2023	Unlimit	93
130	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_4000_5000_cc_CBU_New_Presidential_Edict	200	09-OCT-2023	Unlimit	93
131	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_4000_5000_cc_CKD_New_Presidential_Edict	200	09-OCT-2023	Unlimit	93
132	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_4000_5000_cc_IDK_New_Presidential_Edict	200	09-OCT-2023	Unlimit	93
133	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_5000_cc_or_more_CBU_New_Presidential_Edict	220	09-OCT-2023	Unlimit	93

134	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_5000_cc_or_more_CKD_New_Presidential_Edict	220	09-OCT-2023	Unlimit	93
135	1. Transportation vehicles like: cars, jeeps, vans, four-door or more pickups. Gas- or diesel-powered (Presidential Edict 003 (2023) New Excise Tax)	With_engine_volume_of_5000_cc_or_more_IKD_New_Presidential_Edict	220	09-OCT-2023	Unlimit	93
136	2.1	Alcohol_or_alcoholic_beverages_over_23_degrees_new	110	09-OCT-2023	Unlimit	93
137	2.2	Alcohol_or_alcoholic_beverages_between_23_and_10_degrees_new	90	09-OCT-2023	Unlimit	93
138	2.3	Alcohol_or_alcoholic_beverages_below_10_degrees_new	72	09-OCT-2023	Unlimit	93
139	3.1	Beer_with_alcoholic_beverages_over_5_degrees_new	70	09-OCT-2023	Unlimit	93
140	3.1	Beer_with_alcoholic_beverages_between_0_5_and_5_degrees_new	60	09-OCT-2023	Unlimit	93
141	4.1	Cigars_packed_or_cased_tobacco_and_other_types_of_tobacco_new	72	09-OCT-2023	Unlimit	93
142	4.2	Shredded_tobacco_new	47	09-OCT-2023	Unlimit	93
143	5.1	soft_drinks_sugarycarbonated_and_similar_beverages_new	12	09-OCT-2023	Unlimit	93
144	5.2	energy_drinks_new	17	09-OCT-2023	Unlimit	93

Table 5.2.3.6

In which: Tax Provision / TE: Currently, according to experts' data, it is not enough to accurately classify each provision for each type of goods. We temporarily set = 93 (default is Other provision Excise Tax). It also has the same meaning as "Undefined" or "Unclassified".

In addition, for excise tax, there is an additional decision on adjusting the tax rate for some items. Details are reflected in the document "Presidential Edict 003 (2023) New Excise Tax.pdf". Therefore, in the data table above, there are some new variables with the suffix "_new", which implies the determination of additional tax rates according to the new regulations.

For special consumption tax: Due to the large number of commodity classifications (about 140 items), we designed a system to store them in a data table to minimize the number of columns displayed in tax laws; at the same time, it also helps to expand/add new commodity classification categories more flexibly, without being limited by the number of variables. Detail as described on **Table 5.2.3.6**

5.4.3.4 Create template file and build function on Dashboard of TaxETS system.

Below is some functional interface screens related to **Excise tax**

1. Import and manage Excise Tax

As described above. We use the commodity classification number with the display label "**Variable name / Type of the goods**"

TaxETS

System >

Data Dictionary >

Get Tax Data by Import from Excel >

Data Requirements to identify the type of Repository >

Data Requirements to estimate TE >

- Profit Tax
- Excise Tax Actually**
- Custom Tax Actually
- Individual Tax
- Salary Tax
- Domestic VAT
- For SEZ Developers
- For SEZ Investors

Excise Tax_actually

Dashboard » Excise Tax_actually

Browse Data (Show 1 ~ 8 data from total 8 records)

Search... Add New Import

UNIQ ID FOR ALL RECORD IN SYSTEM	TIN OF ORGANIZATION / PERSON	TE EXCISE PROVISION	DATE TIME APPEAR INCOME (DATE TIME SALE/BUY...)	TOTAL VALUE INCOME	TAX RATE (IF HAVE)	EXCISE TE VALUE FOR COMPARE (IF HAVE)	VARIABLE NAME	ACTION
1468544	45554755-635	11	24/10/2022 10:43:00	1,258,671,600			Beer	
99999	999999		01/01/2021 00:00:00	50,000,000			Casino_services_poker_machines_and_all_kinds...	
11223	509999888-088	12	01/02/0024 00:00:00	350,000,000	10.00%	10.00	Beer	
2281	509999999-001	12	01/02/2019 00:00:00	2,000,000	10.00%	10.00	Agriculture	
158895	158895	123	10/10/2025 00:00:00	10,000	10.00%		Refrigerated_vans_trucks_transporting_liquids...	

MS Excel template file as described in **Table 5.2.3.5** Define the Minimum Data table. Users can download from the Import function interface.

2. Run Benchmark Excise Tax

TaxETS

System >

Data Dictionary >

Get Tax Data by Import from Excel >

Get Tax Data by API >

Run Tax Expenditure ▾

- Run Profit Tax Expenditure
- Run Individual Tax Expenditure
- Run Excise Tax Expenditure**
- Run VAT Tax Expenditure
- Run Custom Tax Expenditure

Report >

Run Excise Tax Expenditure

TIN:

Start Date *:

End Date *:

☐	UNIQU ID FOR ALL RECORD IN SYSTEM	TIN	INPUT DATE	TOTAL EXCISE INCOME	EXCISE TAX RATE	TE VALUE	CURRENCY
☐	123	909954240-900	04/11/2021 05:05:00	1,000,000	10%		LAK
☐	111	123	30/10/2021 15:08:15	1,000	10%		LAK
☐	228	509999999-001	01/02/2021 07:00:00	2,000,000	10%	10	LAK
☐	158895	158895	10/10/2025 07:00:00	10,000	10%		LAK
☐	2281	509999999-001	01/02/2019 07:00:00	2,000,000	10%	10	LAK
☐	11223	509999888-088	01/02/24 07:06:30	350,000,000	10%	10	LAK
☐	99999	999999	01/01/2021 07:00:00	50,000,000			LAK
☐	1468544	45554755-635	24/10/2022 17:43:00	1,258,671,600			LAK

Input:

- TIN
- Start date and End date: Date time when you want to start running Benchmark tax calculations for the selected TIN (that means from time - to time)

Output: The system will automatically search for entered tax data that meets the above conditions and calculate the standard tax value (if met).

2. View Result Benchmark Excise Tax

Run Excise Tax Expenditure Result Data												
Dashboard » Run Excise Tax Expenditure Result Data												
Browse Data												
(Show 1 ~ 5 data from total 5 records)												
Search...												
UNIQ ID FOR ALL RECORD IN SYSTEM	TIN OF ORGANIZATION / PERSON	TE EXCISE PROVISION	TOTAL VALUE INCOME	TAX RATE	EXCISE TE VALUE FOR COMPARE	DATE TIME APPEAR	AMOUNT OF EXCISE COLLECTED/PAID	EXCISE TE	DATETIME CREATED	USER CREATE/UPDATE	NOTE	ACTION
1468544	45554755-635	93	1,258,671,600	50.00%	629,335,800	24/10/2022 10:43:00	0	629,335,800	18/11/2025 10:46:19			
11223	509999888-088	93	350,000,000	50.00%	175,000,000	01/02/2023 00:00:00	5,000,000	170,000,000	13/11/2025 15:40:50			
228	509999999-001	93	2,000,000	25.00%	500,000	01/02/2021 00:00:00	5,000	495,000	13/11/2025 15:34:45			
123	909954240-900	93	12,000,000	35.00%	4,200,000	03/11/2021 22:05:00	5,000	4,195,000	10/11/2025 14:49:41			
111	123	93	1,000	30.00%	300	30/10/2021 22:05:00	1,000	-700	09/11/2025 16:40:53			
Items per page: 10												

Output:

- + Tax rate
- + Excise TE value for compare (Excise Tax benchmark value)
- + Amount of Excise tax collected/PAID (if have)
- + Excise Tax TE (TE = Excise Tax Benchmark - Excise Tax PAID)
- + Datetime RUN
- + User ID RUN

5.4.3.5 Test Result.

Below is the result from testing of TaxETS for example a product: Beer.

Run Excise Tax Expenditure Result Data Dashboard » Run Excise Tax Expenditure Res

Browse Data
(Show 1 ~ 1 data from total 1 records)

580761167-900

UNIQ ID FOR ALL RECORD IN SYSTEM	TIN OF ORGANIZATION / PERSON	TE EXCISE PROVISION	TOTAL VALUE INCOME	TAX RATE	EXCISE TE VALUE FOR COMPARE	DATE TIME APPEAR	AMOUNT OF EXCISE COLLECTED/PAID	EXCISE TE
4523	580761167-900	93	357,300	50.00%	178,650	20/07/2023 13:30:00	0	178,650

And

5	ເຫຼົ່າ ຫຼື ເຄື່ອງດື່ມ ທີ່ມີທາດເຫຼົ່າ	
	ເຫຼົ່າ ຫຼື ເຄື່ອງດື່ມ ທີ່ມີທາດເຫຼົ່າ ເກີນ 23 ສ່ວນຮ້ອຍ	70%
	ເຫຼົ່າ ຫຼື ເຄື່ອງດື່ມ ທີ່ມີທາດເຫຼົ່າ ແຕ່ 23 ສ່ວນຮ້ອຍ ລົງມາ	60%
6	ເບຍ	50%
7	ປະເພດຢາສູບ	
	ຢາຊີກາ, ຢາສູບເປັນກອກ ຫຼື ເປັນຊອງ ແລະ ຢາສູບປະເພດອື່ນ	50%
	ຢາສູບໄຟຟ້າ, ນ້ຳຢາທີ່ມີສ່ວນປະສົມຂອງນິໂກຕິນ ສຳລັບໃຊ້ກັບຢາສູບໄຟຟ້າ	60%
	ຢາເສັ້ນ	35%

ຂໍ້ມູນຕົວຢ່າງອ້າງອີງຈາກເວັບໄຊ : <https://laotradeportal.gov.la/lo-la/site/display/1832>

5.4.4 Customs Tax

5.4.4.1 Customs Tax calculation rule prototype and legal documents

- Legal documents:

- + Customs Law 81 (2020)_English.pdf
- + MS Excel file containing HS CODE table content and ASEAN regional cooperation program “For print AHTN 2017 + NOR + WTO + ATIGA + AC+ AI+AANZ+AJ+AK+APTA+LV LATEST 12.9.19 .xlsx”

- **Provision list provided by MOF expert (from MS Excel file)**

TE #	Type	Legal Basis	TE Description
55	Exemption	IPL Art. 12.1 (amendment)	Import of materials, equipment which may not be supplied or produced in the Lao PDR, to form the fixed assets, and of machinery and vehicles directly used for production.
56	Exemption	IPL Art. 12.2 (amendment)	Import of raw materials, equipment, mineral ore and parts to be used in the production for export.

Table 5.2.4.1

- **Customs Tax Rate Summary Table**

Tax Unit	Tax Base	Tax Period	Tax Rate
Importer	CIF value of imported goods	At the time of import	Benchmark duty rates are those described in the ASEAN Harmonized Tariff Nomenclature, incorporating customs agreements and free trade areas*. Different rates form the benchmark tax rates, depending on the product category (at the HS Code level). TE arise where some imports are charged lower or zero tariffs for goods that ought to be taxed under the appropriate customs agreement.

Table 5.2.4.2

5.4.4.2 Define the Minimum Data table

As we analyzed at the beginning of this section, the minimum data on Excise Tax should include the following information:

No.	Column Name	Description	Calculation/Input/Remark
1	Uniq ID for all Record in system	ID Transactions in system	Mandatory , Unique Identify
2	TIN of Organization / person	Tax Identification Number of the enterprise.	Mandatory
3	TE Customs provision (if have)	Provision of this Tax transaction	Option Only for double check
4	Date time appear Income (date time sale/buy..)	The time of transaction of goods/services that may generate tax.	Mandatory
5	Total value income	Total Amount of the goods... to be taxed	Mandatory
6	Tax rate (if have)	Tax rate	Option Only for double check
7	Customs TE value for compare (if have)	TE Value from other source...	Option Only for double check
8	Currency type/name, default is LAK	Currency	Option
9	Amount of Excise collected/PAID	The amount the taxpayer paid for this tax	Option to Calculate TE Value Remaining
10	Customs Tax estimate by Rules	TE Value benchmark from other source...	Option Only for double check
11	Variables to classification kind of Customs tax (Type of the goods) For customs duties. In addition to classifying goods, we also need to determine which cooperation program the goods are exported/imported under in the region to find the appropriate tax rate. In this context, we divide into 2 sub-variables "HS Code Basic / AHTN 2017 CODES" and "Belong Program"	Variables are used to classify goods and thereby find the tax rate that will be applied. Please see the table below for details on the values of this variable. Is the HS CODE managed by	Mandatory. Select a value from the list below to classify the goods/services.... that are subject to this tax.

	HS Code Basic / AHTN 2017 CODES	the current customs system. All taxable goods must have this corresponding code.
	Belong Program	Determine which goods will be taxed under which cooperation program

Table 5.2.4.3

5.4.4.3 Inherit data content from MS Excel files provided by MOF experts

We received a sample data file for reference from the MOF expert group. Data file with name “Customs, Excise, Import VAT Test.xlsx” with the data structure described as follows:

Province	TIN	No	BorderCode	Border	Type of Customs Declaration	Process	F	Special Role	Import Document Number	Import Document Date	Assess Number	Assess Date	TryD Receipt No	TryD Receipt Date	Import/Export Name
Khammouan	580761167-900	3335	R3C60	ດ່ານພານີສາກົນ	IM	4000	480		10158	24/10/2022	10121	24/10/2022	221024000105	24/10/2022	ບ/ສ ຊີໂນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-900	1597	R1C10	ດ່ານພານີສາກົນ	IM	4000	481		30686	17/09/2021	30658	18/09/2021	20210918490412	18/09/2021	ບ/ສ ຊີໂນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-900	5556	R1C10	ດ່ານພານີສາກົນ	IM	4000	441		27812	26/06/2023	27559	26/06/2023	230626000412	26/06/2023	ບ/ສ ຊີໂນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-900	6979	R1C10	ດ່ານພານີສາກົນ	IM	4000	000	ACN	32254	20/07/2023	31880	20/07/2023	230720000284	20/07/2023	ບ/ສ ຊີໂນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-900	4918	R1C10	ດ່ານພານີສາກົນ	IM	4000	441		59925	28/11/2023	59642	29/11/2023	231129000065	29/11/2023	ບ/ສ ຊີໂນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ
Khammouan	580761167-900	2132	R1C10	ດ່ານພານີສາກົນ	IM	4000	481		22068	08/07/2021	21992	08/07/2021	202107081456624	08/07/2021	ບ/ສ ຊີໂນ-ອາກຣີໂປຕາສ ຈຳກັດໂພນພະເນົາໄຊເສດຖ

Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF
CustomsDeclareTIN	CustomsDeclareName	ExportCountryCode	Destinationcountrycode	OriginalCountryCode	List	GoodsHS	GoodsDescription	Quantity	Unit	DeclareWeight	ActualWeight	InvoiceAmountUSD	InvoiceAmountLAK	Customs	Excise
977704808900	ບສ ເອັກຊິມ ຈຳກັດຜູ້ດຽວໄຊເສດຖາ, ນຳ	VN	LA	VN	1	3.924E+09	ຖ້ຳຜະລິດຕະພັນ(ຂະໜາດໂຕນ:90*90*125ລຸມ) ຈຳນວນ	12700	KGM	26774	26670	76200	1258671600	0	
625607046000	ບ/ສ ຮຸ່ງອາລຸນການຄ້າ ຂາອອກ-ຂາເຂົ້າບ	CN	LA	CN	1	3923909000	ຖ້ຳຜະລິດຕະພັນ(ຂະໜາດໂຕນ) 415໒ປາ=8,300ອັນໃນ	15770	KGM	16060	15770	49800	477831000	0	
681630811000	ບໍລິສັດ ຂົນສົ່ງສາກົນ ຈຳກັດຈອມແກ້ວ, ນ	CN	LA	CN	1	8.426E+09	ເຄື່ອງໝັກໜັກໄຟຟ້າແບບຄານດຽວ LB5T-16.5Mພ້ອມ	3	NAR	31470	31134	45000	800775000	0	
681630811000	ບໍລິສັດ ຂົນສົ່ງສາກົນ ຈຳກັດຈອມແກ້ວ, ນ	CN	LA	CN	6	8.431E+09	ປອກຈັບດ້ານລຸ່ມ =12ອັນ=1ແກ້ວ=68ກິໂລ	12	NAR	70	68	170	3247340	0	
681630811000		CN	LA	CN	2	7.216E+09	ເຫລັກຮ່ອງ 160*63*6mm,6M =72ອັນ=7.400ກິໂລ	7400	KGM	7400	7400	18500	384707500	0	
625607046000	ບ/ສ ຮຸ່ງອາລຸນການຄ້າ ຂາອອກ-ຂາເຂົ້າບ	CN	LA	CN	4	7208100000	ເຫຼັກແຜ່ນລາຍ = 1,077ກິໂລ	1077	KGM	1077	1077	862	8176932	0	

AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT
Excise	VAT	Profit	RoadFund	Total	AJ	Exempt_Customs	Exempt_Excise	Exempt_VAT	TE_Customs	TE_Excise	TE_VAT	Provision_Customs	Provision_Excise	Provision_VAT
0	0	0	0	0	Assessed	251734400	0	105728500	251734400		105728500	4000-480		4000-480
0	0	0	0	0	Assessed	95566200	0	57339800	95566200		57339800	4000-481		4000-481
0	0	0	0	0	Assessed	40038800	0	58857000	40038800		58857000	4000-441		4000-441
0	357300	252400	0	609700	Assessed	162400	0	0	162400			4000-000		
0	0	0	0	0	Assessed	19235400	0	28276100	19235400		28276100	4000-441		4000-441
0	0	0	0	0	Assessed	408900	0	858600	408900		858600	4000-481		4000-481

Data structure in MS Excel file		Matching results (The results match what we need)	
No	Column name	Compatible / or usable or not	Reason why NOT
1	Province	YES	
2	TIN	YES	
3	No	YES (use for ID_Transactions)	

4	BorderCode	NO	less relevant / redundant / used for other purposes
5	Border	NO	
6	TpyeofCustomsDeclare	NO	
7	Processtp	NO	
8	F	NO	
9	SpecialRole	NO	
10	ImportDocumentNumber	NO	
11	ImportDocumentDate	YES (~ Input_date)	
12	AssessNumber	NO	less relevant / redundant / used for other purposes
13	AssessDate	NO	
14	TryDReceiptNo	NO	
15	TryDReceiptDate	NO	
16	ImportExportName	NO	
17	CustomsDeclareTIN	NO	
18	CustomsDeclareName	NO	
19	ExportCountryCode	NO	
20	Destinationcountrycode	NO	
21	OriginalCountryCode	NO	
22	List	NO	
23	GoodsHS	YES (~ Total Income Value)	but use for HS CODE in Customs Tax)
24	GoodsDescription	NO	less relevant / redundant
25	Quantity	NO	
26	Unit	NO	
27	DeclareWeight	NO	
28	ActualWeight	NO	
29	InvoiceAmountUSD	NO	
30	InvoiceAmountLAK	YES (~ Total Income Value)	
31	Customs	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
32	Excise	NO	As a result for tax VAT, only use for reference/check in other scope
33	VAT	NO	
34	Profit	NO	less relevant / redundant / only use for reference/check in other scope
35	RoadFund	NO	
36	Total	NO	
37	AJ	NO	

38	Exempt_Customs	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
39	Exempt_Excise	NO	As a result for tax Excise, only use for reference/check in other scope
40	Exempt_VAT	NO	As a result for tax VAT, only use for reference/check in other scope
41	TE_Customs	YES (~ use for reference/DOUBLE check AFTER run benchmark)	
42	TE_Excise	NO	As a result for tax Excise, only use for reference/check in other scope
43	TE_VAT	NO	As a result for tax VAT, only use for reference/check in other scope
44	Provision_Customs	NO	We do not have a code table corresponding to "Provision_Customs", "Provision_Excise", "Provision_VAT", and have not been provided with a classification structure according to this code table.
45	Provision_Excise	NO	
46	Provision_VAT	NO	

Table 5.2.4.4

Note: The data highlighted in yellow is useful for VAT tax calculation and we will use it in the calculation formula or double check result benchmark.

After careful consideration of each case and with an eye to future expansion, we have determined that the minimum data sheet (including both mandatory and optional information) should include at least the following information:

No.	Column Name	Description	Calculation/Input/Remark	The data provided by the experts is of similar meaning/usability.
1	Uniq ID for all Record in system	ID Transactions in system	Mandatory , Unique Identify	No
2	TIN of Organization / person	Tax Identification Number of the enterprise.	Mandatory	TIN

3	TE Customs provision (if have)	Provision of this Tax transaction	Option Only for double check	
4	Date time appear Income (date time sale/buy..)	The time of transaction of goods/services that may generate tax.	Mandatory	ImportDocumentDate
5	Total value income	Total Amount of goods/services ... to be taxed	Mandatory	InvoiceAmountLAK or InvoiceAmountUSD
6	Tax rate (if have)	Tax rate	Option Only for double check	
7	Customs TE value for compare (if have)	TE Value from other source...	Option Only for double check	Exempt_ Customs
8	Currency type/name, default is LAK	Currency	Option	
9	Amount of Excise collected/PAID	The amount the taxpayer paid for this tax	Option to Calculate TE Value Remaining	
10	Customs Tax estimate by Rules	TE Value benchmark from other source...	Option Only for double check	TE_ Customs
11	Variables to classify Excise tax (Type of the goods) HS Code Basic / AHTN 2017 CODES Belong Program	Variables are used to classify goods and thereby find the tax rate that will be applied. Please see the table below for details on the values of this variable.	Mandatory. Select a value from the list below to classify the goods/services.... that are subject to this tax.	GoodsHS Missing

Table 5.2.4.5

Conclusion: For Custom Tax, the same as for other Tax that related to type of product, the data provided by the expert (or import template) has a bit discrepancy in comparison to our system design. Expert data relies on the end result, like Custom Tax Paid but less concerns about the detail of goods detail, we believe that the data is based on ASYCUDA system, that has made this calculation already. In the other hand our system is design so, that having information to classify taxable goods/services, we can calculate Custom Tax Paid according to related provision. Nervelessness both method provide the same result, while the expert method is kind of shortcut and rely on the data from external system such as ASYCUDA.

This is also the MS Excel file structure used in the Excise tax data import function.

Based on the "Customs Tax Rate Summary Table" and Provision list provided by MOF expert (from MS Excel file) above, which is provided by the expert, we have built specific variables, corresponding to the rules / provisions as follows:

ID Type	Belong Program
1	Normal
2	MFN (WTO)
3	ATIGA
4	ACFTA
5	AKFTA
6	AJCEP
7	AANZFTA
8	AIFTA
9	APTA
10	Lao-Viet

Table 5.2.4.6

With the tax rate information, we got from the Excel file "*For print AHTN 2017 + NOR + WTO + ATIGA + AC+AI+AANZ+AJ+AK+APTA+LV LATEST 12.9.19 .xlsx*", we designed a data table to store them including product code (HS CODE), description, participating program and corresponding tax rate as follows:

This data table contains information on the tariff rates of approximately 140 thousand items participating in 10 cooperation programs, each program has a shortest term of 1 year, the longest is indefinite term. Due to the large volume of data, we only list a few for illustration as follows:

HS_CODE (AHTN 2017 CODES)	Name lao	Name eng	Unit	Tax rate value	Belong_ program	Effective_date_fro m	Effective_date_to
8414802900	- - - - ອື່ນໆ	---- Other	ອັນ	0	ACFTA	1/1/2018	12/31/2018
8414803000	- - ບໍ່ ມີ ລູກ ສູບ ສໍາ ລັບ ກັ່ງ ຫັນ ແກ້ ສ	-- Free-piston generators for gas turbines	ອັນ	0	ACFTA	1/1/2018	12/31/2018
8414804100	- - - ເຄື່ອງວັດ ແທກ ຄວາມ ດັນ ຂອງ ແກ້ ສ ເໝາະສໍາລັບ ໃຊ້ ໃນ ການ ຊຸດ ດເຈາະ ນໍ້າ ມັນ	--- Gas compression modules suitable for use in oil drilling operations	ອັນ	0	ACFTA	1/1/2018	12/31/2018
8414804200	--- ຊະນິດທີ່ໃຊ້ສໍາລັບແອ ວົດຍົນ	--- of a kind used for automotive air conditioners	ເຄື່ອງ	0	ACFTA	1/1/2018	12/31/2018
0101900000	- ອື່ນໆ	- Other	ໂຕ	10	Normal	1/1/1970	12/31/3000
0102210000	- ສາຍພັນບໍ່ລົດສຸດສໍາ ລັບ ເຮັດແນວ ພັນ	-- Pure-bred breeding animals	ໂຕ	5	Normal	1/1/1970	12/31/3000
0102291100	---- ງົວຕອນ	---- Oxen	ໂຕ	10	Normal	1/1/1970	12/31/3000
0102291900	---- ອື່ນໆ	---- Other	ໂຕ	10	Normal	1/1/1970	12/31/3000

0102299000	--- ອື່ນໆ	--- Other	ໂຕ	10	Normal	1/1/1970	12/31/3000
0102310000	- ສາຍພັນບໍ່ວິສຸດສຳ ລັບ ເຮັດແນວ ພັນ	-- Pure-bred breeding animals	ໂຕ	10	Normal	1/1/1970	12/31/3000

This structure ensures that the entire tax rate data table of many items can be stored and applied for many years, participating in many cooperation programs; at the same time, allowing flexible and quick amendments and additions. All data has been standardized and entered the TaxETS system. We have attached this report as an MS Excel file (File name: **tbl_Cat_Custom_AHTN2017_new.xlsx**)

Below is the initial input data table (From the data file received from MOF expert)

AHTN 2017 CODES		ລາຍການສັນຄ້າ	DESCRIPTIONS	Unit	Normal	MFN (WTO)	ATIGA	ACFTA			AKFTA							AJCEP					
				ຫົວໜ່ວຍປະຕິບັດ		2018~	2018~	2018	2019	2020	###	###	###	###	###	###	###	2019-20	2021-22	2023-25	2026~	2011	
8704.22.39		----- ອື່ນໆ	----- Other:																				
8704.22.39	10	----- ທີ່ມີນ້ຳໜັກລວມເກີນ 6 ໂຕນ ແຕ່ບໍ່ເກີນ 15 ໂຕນ	----- g.w.v exceeding 6 t but not exceeding 15 t	ຕົ້ນ	30	15	0	0	0	0	20	20	20	20	20	20	10	10	5	0	20		
8704.22.39	90	----- ທີ່ມີນ້ຳໜັກລວມເກີນ 15 ໂຕນ ແຕ່ບໍ່ເກີນ 20 ໂຕນ	----- g.w.v exceeding 15 t but not exceeding 20 t	ຕົ້ນ	30	15	0	0	0	0	20	20	20	20	20	20	10	10	5	0	20		
		----- ອື່ນໆ	----- Other :																				
8704.22.41	00	----- ລົດຕັ້ງເຢັນ	----- Refrigerated lorries (trucks)	ຕົ້ນ	30	15	0	0	0	0	20	20	20	20	20	20	10	10	5	0	20		
8704.22.42	00	----- ລົດເກັບຂີ້ເຍື້ອ ທີ່ມີອະນາຄອນອັດຂີ້ເຍື້ອຕິດຢູ່ນຳ	----- Refuse/garbage collection vehicles having a refuse	ຕົ້ນ	30	15	0	0	0	0	20	20	20	20	20	20	10	10	5	0	20		

5.4.4.4 Create template file and build function on Dashboard of TaxETS system.

Below is some functional interface screens related to **Excise tax**

1. Import and manage Customs tax

As described above. We use the commodity classification number with the display label "HS Code Basic / AHTN 2017 CODES" and "Belong Program"

System >

Data Dictionary >

Get Tax Data by Import from Excel >

Data Requirements to identify the type of Repository >

Data Requirements to estimate TE >

Profit Tax

Excise Tax Actually

Custom Tax Actually

Individual Tax

Salary Tax

Domestic VAT

For SEZ Developers

For SEZ Investors

Customs, Excise Tax, Import VAT from ASYCUDA

Non-Tax Land

Custom Tax Actually

Browse Data

(Show 1 ~ 20 data from total 20 records)

Search...

UNIQ ID FOR ALL RECORD IN SYSTEM ↑↓	TIN OF ORGANIZATION / PERSON ↑↓	TE CUSTOM PROVISION ↑↓	DATE TIME APPEAR INCOME (DATE TIME SALE/BUY...) ↑↓	TOTAL VALUE INCOME ↑↓	TAX RATE (IF HAVE) ↑↓	CUSTOM TE VALUE FOR COMPARE (IF HAVE) ↑↓	HS_CODE (IF HAVE) ↑↓	BELONG PROGRAM/PROJECT ↑↓
1460	14564578-124	11	24/10/2023 10:04:00	251,734,400	10.00%	0.00	3923909000	AJCEP
9	333		26/06/2023 00:00:00	800,775,000			8426110000	Lao-Viet
8	222		17/09/2021 00:00:00	477,831,000			3923909000	Normal
7	111		24/10/2022 00:00:00	1,258,671,600			3923909000	Normal
6	580761167-900		11/08/2023 00:00:00	8,176,932			7208100000	Normal
5	580761167-900		28/11/2023 00:00:00	384,707,500			7216100000	Normal
4	580761167-900		20/07/2023 00:00:00	3,247,340			8431499000	Normal

MS Excel template file as described in **Table 5.2.4.5** Define the Minimum Data table. Users can download from the Import function interface.

2. Run Customs Tax Benchmark

System >

Data Dictionary >

Get Tax Data by Import from Excel >

Get Tax Data by API >

Run Tax Expenditure >

Run Profit Tax Expenditure

Run Individual Tax Expenditure

Run Excise Tax Expenditure

Run VAT Tax Expenditure

Run Custom Tax Expenditure

Report >

Run Custom Tax Expenditure

TIN

Start Date *

End Date *

Enter TIN

mm/dd/yyyy

mm/dd/yyyy

View data

Search

Run

☐	UNIQ ID FOR ALL RECORD IN SYSTEM	TIN	INPUT DATE	TOTAL CUSTOM INCOME	CUSTOM TAX RATE	TE VALUE	CURRENCY
☐	123	12345	04/11/2021 04:29:00	10,000,000	10%		LAK
☐	6667	666	01/01/2021 07:00:00	300,000,000			LAK
☐	11122	1232323	01/01/2025 19:00:33	10,000	10%		LAK
☐	1283	580761167-900	20/07/2023 07:00:00	3,247,340	10%		LAK
☐	12834	580761167-111	20/07/2023 07:00:00	251,734,400			LAK
☐	12835	580761167-122	20/07/2021 07:00:00	251,734,400	10%		LAK
☐	14235	145266647-333	20/07/2023 07:00:00	162,400			LAK
☐	14236	145266647-333	20/07/2023 07:00:00	162,400			LAK

Input:

- TIN
- Start date and End date: Date time when you want to start running Benchmark tax calculations for the selected TIN (that means from time - to time)

Output: The system will automatically search for entered tax data that meets the above conditions and calculate the standard tax value (if met).

2. View Result Customs Tax Benchmark

Run Custom Tax Expenditure Result Data

Dashboard » Run Custom Tax Expenditure Result Data

Browse Data

(Show 1 ~ 17 data from total 17 records)

Search...

Add New

UNIQ ID FOR ALL RECORD IN SYSTEM	TIN OF ORGANIZATION / PERSON	TE CUSTOM PROVISION	TOTAL VALUE INCOME	TAX RATE	CUSTOM TE VALUE FOR COMPARE	DATE TIME APPEAR	AMOUNT OF CUSTOM COLLECTED/PAID	CUSTOM TE	DATETIME CREATED	USER CREATE/UPDATE	NOTE	ACTION
1460	14564578-124	94	251,734,400	5.00%	12,586,720	24/10/2023 10:04:00	0	12,586,720	18/11/2025 14:56:37			
6	580761167-900	94	8,176,932	5.00%	408,847	11/08/2023 00:00:00	0	408,847	18/11/2025 09:14:32			
5	580761167-900	94	384,707,500	5.00%	19,235,375	28/11/2023 00:00:00	0	19,235,375	18/11/2025 09:14:32			
4	580761167-900	94	3,247,340	5.00%	162,367	20/07/2023 00:00:00	0	162,367	18/11/2025 09:14:32			
3	580761167-900	94	800,775,000	5.00%	40,038,750	26/06/2023 00:00:00	0	40,038,750	18/11/2025 09:14:32			
2	580761167-900	94	477,831,000	20.00%	95,566,200	17/09/2021 00:00:00	0	95,566,200	18/11/2025 09:14:32			

Output:

- + Tax rate
- + Custom TE value for compare (Custom Tax Benchmark)
- + Amount of Custom Tax collected/PAID (if have)
- + Custom Tax TE (TE = Custom Tax Benchmark - Custom Tax PAID)
- + Datetime RUN
- + User ID RUN

5.4.4.5 Test Result.

For the test we use the product with the code 3923.90.90.00 and belong Program/Project with the type Normal, the Custom Tax Rate of which 20% according to MOF regulation.

HS_CODE (if have)

3923909000

Input HS Code basic

Belong Program/Project

Normal

Input HS Code basic

Run Custom Tax Expenditure Result Data

Dashboard » Run Custom Tax Expenditure Result Data

Browse Data

(Show 1 ~ 1 data from total 1 records)

14564578-124

+ Add New

UNIQ ID FOR ALL RECORD IN SYSTEM ↕	TIN OF ORGANIZATION / PERSON ↕	TE CUSTOM PROVISION ↕	TOTAL VALUE INCOME ↕	TAX RATE ↕	CUSTOM TE VALUE FOR COMPARE ↕	DATE TIME APPEAR ↕	AMOUNT OF CUSTOM COLLECTED/PAID ↕	CUSTOM TE ↕
1460	14564578-124	94	251,734,400	20.00%	50,346,880	24/10/2023 10:04:00	0	50,346,880

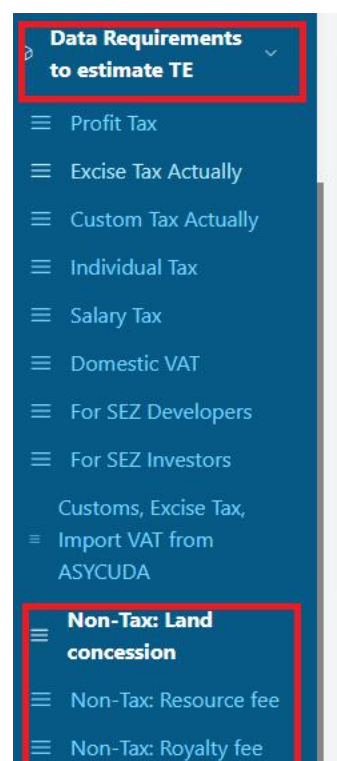
Based on regulation from MOF:

AHTN 2017 CODES	ລາຍການສິນຄ້າ	DESCRIPTIONS	Unit	Normal	MFN (WTO)	ATIGA	ACFTA		
			ຫົວໜ່ວຍ	ປົກກະຕິ	2018~	2018~	2018	2019	2020
3923.90	- ອື່ນໆ:	- Other :							
3923.90.10	00 - - ຫອດສໍາລັບໃສ່ຢາສີພັນ	- - Toothpaste tubes	ກລ	20	10	0	0	0	0
3923.90.90	00 - - ອື່ນໆ	- - Other	ກລ	20	10	0	0	0	0

Which proves that the calculation in our system is correct.

5.4.5 NON-Tax

For NON-TAX tax, according to the definition of the Ministry of Finance expert, this is only data for storage and statistical purposes, so there will be no Run Benchmark function group (No Calculation) but only Import and View Report function groups. We have also completed the construction of Import/View/Edit/Delete functions corresponding to the sub-taxes of NON-Tax. This function group can be found in the left menu of the TaxETS system as shown below.



Up to now, we have received sample files of these fees, we have built the function to import them into the system.

We also received a sample of real data from MOF expert and it was imported into the easy to use functional tape system on TaxETS interface.

We received a sample data file on **Non Tax Land Concession** and it contains data on this tax only. We have built the data exactly according to this sample file.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	No.	TIN	District	Province	Tax Item	Year	Confirm Date	Concession areas (ha)	Benchmark rate, USD	Contracted rate, USD	Concession fee paid, USD	Benchmark Value, USD	Non-Tax TE, USD	Non-tax Provision
1														
2	1	734718327-000	Nong	Savannakhet	State land concession fee	2022	3/1/2022	500	12	5	2,500	6,000	3,500	
3														
4														
5														

File name is Non-Tax Test.xlsx

For the remaining two types of fees (**Resource fee** and **Royalty fee**), we understand that the actual data structure to store them may be very different from the design prototype. However, since there is no specific data model yet, we are still using the old prototype that we received from MOF experts in the early stages. If we are provided with new actual data files, we will update them as required by MOF.

Resource fee								Dashboard » Resource fee	
Browse Data								Search... Add New Import	
(Show 1 ~ 1 data from total 1 records)									
NO	TIN	DATE OF INVESTMENT LICENSE	TYPE OF NATURAL RESOURCE	RESOURCE FEE RATE	RESOURCE FEE COLLECTED	YEAR DATA	ACTION		
1	010-00000005	2025-05-09 00:00:00	23	23	34	2025	View Edit Delete		
								Items per page: 10	

Royalty fee Dashboard » Royalty fee

Browse Data Search... Add New Import

(Show 1 – 1 data from total 1 records)

NO ↕	TIN ↕	DATE OF INVESTMENT LICENSE ↕	ELECTRICITY SALE VALUE ↕	ROYALTY FEE RATE ↕	ROYALTY FEE COLLECTED ↕	YEAR DATA ↕	ACTION
2	010-00000005	2025-05-21 00:00:00	23	45	45	2025	👁 ✎ 🗑

Items per page: 10 ▼

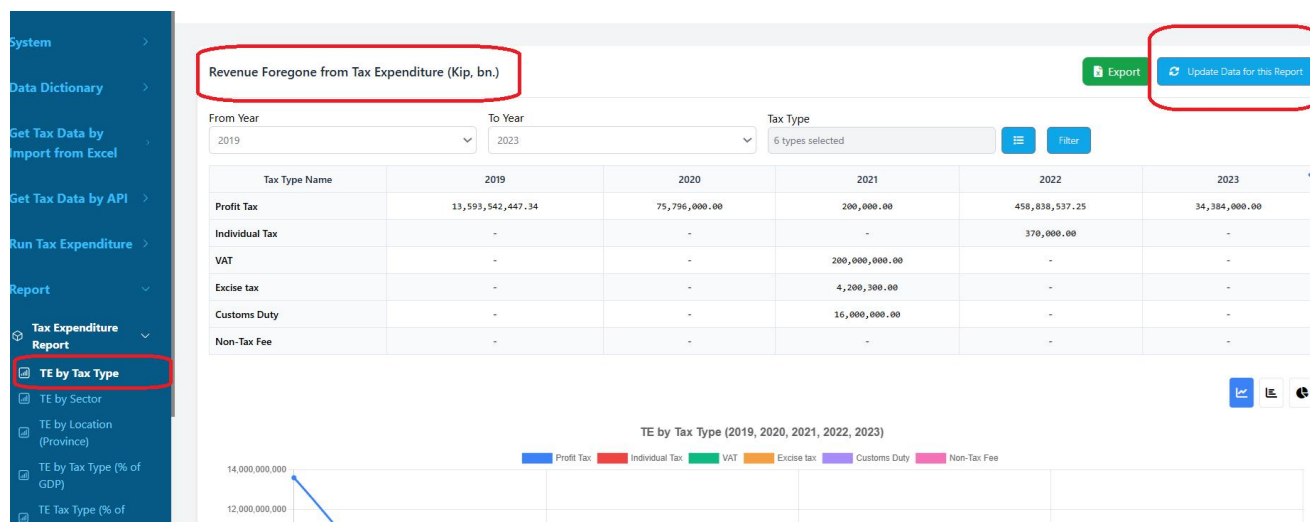
Before concluding this section, we would like to briefly present the report structure and report display mechanism of the taxes that we have built. Up to now, we have built all the reports according to the templates provided by MOF experts. We will continue to update more report templates if there are more requests from MOF and receive specific template files.

The report display process is as follows:

1. Perform the Benchmark tax calculation function for each tax you want to view the report (the purpose is to add newly entered data or update the latest data from other users in the system).
2. Open related reports and use the additional data refresh function on each report interface.

Ex:

Export Update Data for this Report	
Filter	
2022	2023
458,838,537.25	34,384,000.00
370,000.00	-



This process applies to all reports. Its purpose is to ensure that reports are always generated using the most current data available in the system. This is because the TaxETS system is multi-user, so there may be multiple users performing data entry or standard tax functions. And of course, you will always want to view reports using the most current data.

6 Summary of Development and Testing Status

As of 19 November 2025, the Tax Expenditures Tracking System (TaxETS) has reached the following milestones:

1. All UNDP feedback items from Deliverable 3 either implemented or ready for immediate finalisation upon domain registration.
2. Profit Tax TE calculation fully developed, rigorously tested, and validated as 100 % accurate against the MOF Expert's results.
3. Personal Income Tax, VAT, Excise Tax, and Customs Duty modules fully developed with import templates, calculation logic, and reporting functions ready; initial tests show perfect alignment with the Expert's calculations.
4. Non-Tax incentives module completed (import + reporting).
5. Comprehensive HS Code tariff table (~140,000 lines) imported and linked for Customs Duty calculations.
6. Security, logging, backup, and role management significantly strengthened.
7. User interface refined based on actual usage experience.
8. Draft User Manuals (End-User and Administrator versions) prepared and ready for final review.

The system is now ready for intensive real-data testing, user acceptance testing, and training. Remaining tasks (final template polishing, incorporation of any additional sample data, final security hardening, and training material finalisation) are minor and scheduled for completion by mid-December 2025.

7 Conclusion and Way Forward

TaxETS is now a mature, secure, accurate, and highly flexible platform that can serve the Ministry of Finance for decades. It successfully handles both current data availability constraints and future scenarios with complete, detailed data flows from TaxRIS, ASYCUDA, MOIC, and other systems.

We extend our sincere thanks to the Fiscal Policy Department, the Ministry of Finance Expert(s), and UNDP for their unwavering support, rapid feedback, and expert guidance throughout the project. Their contributions have been essential to the system's quality and practical relevance.

The remaining steps are:

1. Final security hardening (domain + SSL + MFA) – awaiting domain registration.
2. Final round of real-data testing with FPD/Exported data (Dec 2025).
3. Completion of User Manuals and training materials.
4. Conduct training sessions (scheduled for December 2025).
5. Official handover and go-live by 31 December 2025.

APIS Company Limited confirms its full commitment to supporting the Ministry of Finance during the warranty period and beyond, ensuring TaxETS continues to evolve in line with Lao PDR's fiscal transparency objectives.

We are confident that TaxETS will become a cornerstone tool for evidence-based tax policy in Lao PDR

Appendix 1: Feedback by UNDP to the Deliverable 3 and APIS Response

Feedback on database development by UNDP

Date: JULY 07, 2025

1. User Management:

- Authentication: Since this database contains sensitive information, maybe better to implement MFA
- Also, when I tried creating a test account, I can put any email (test1@gmail.com) which doesn't belong to me and not sure if it's existed or not, maybe better to have email verification too
- I found that the weak password, such as '123456', can be set without restriction. Combination of letter(s) and symbol(s) are recommended.
- 'STAFF' account can create a user account, maybe restrict this function to only 'Super Admin'

2. Role Management:

- Should have a clear guideline of who can create/read/update/delete which functions
- We would like to review the manuals to be developed by APIS.

3. Under the menu: Get Tax Data by importing from Excel and by API:

- When login with other account types (except Super Admin), these functions do not work. I assume they are intended to be handled by the 'Super Admin' ? (please correct me if I'm wrong).
- If that's the case, these should probably be hidden for other accounts. The same should apply to any other functions that are not designated for certain account types.

4. Under the menu: Get Tax Data by Import from Excel -> Data Requirements to estimate TE -> Profit Tax

- There's no option to edit/delete the specific line, as well as batch deletion feature for handling large amounts of data
- Maybe the 'TIN' information needs to be updated.

5. User activity tracking: edits, deletions, submissions, etc.

6. Admin tracking and monitoring:

- Error log viewer (database error, etc.)

- Users logs for monitoring users transactions in the system, for security reasons and backup if any changes are made without intention. This should be monitored by the Admin users.
- Export logs to CSV, PDF, etc.

7. Backup and Restore:

- In case a user accidentally deletes data or for other purposes.
- Prepared for sustainability of data available for the long term when data bandwidth might increased in the system and causes a slow searching/responding.

Feedback on database development by APIS

Please see our feedback highlighted in blue.

1. User Management:

- Authentication: Since this database contains sensitive information, maybe better to implement MFA

***Our solution:** As of now, we are still in the process of development and refinement, to better serve the testing in multiple methods, so we are storing data in plaintext (unencrypted). We have a plan for this, after all the functions are completed and ready to go live, we will switch all data to encrypted mode. As you know, in the development plan of the project, there is a process of "Test and improve the database". Accordingly, we will release version 3.0 or 4.0. And in these versions, we will update according to your comments.*

*→ **STATUS: DONE** (Our proposal is: we will store encrypted versions of text data related to business/company information. We will still store other data in plaintext/unencrypted form. This method will avoid overloading the system when the system has to continuously decrypt data that is numeric, less sensitive). You and your team may have any additional suggestions that you consider reasonable in this matter.*

*The system is currently under development so temporarily use IP address for access. We also have plans to apply SSL security certificates to this system, but that can only be done after you have registered an official **domain name** for the system. After having an official domain name, we will apply SSL security certificates. Combined with data encryption, all your data will be safe.*

→ **STATUS: *Waiting to register domain and SSL.*** *Our suggestion is that you should register a specific domain name for this system, and also register to use the security certificate provided by service providers in Laos. You will have to pay a small annual fee for this work. You can authorize APIS to do this for you.*

- Also, when I tried creating a test account, I can put any email (test1@gmail.com) which doesn't belong to me and not sure if it's existed or not, maybe better to have email verification too

Our solution: Yes, We will update your comment in version 3.0

→ **STATUS: *In progress*** *We are checking and will update you with details. In case SSL is required, we will complete it after registering the domain and applying the SSL certificate.*

- I found that the weak password, such as '123456', can be set without restriction. Combination of letter(s) and symbol(s) are recommended.

Our solution: Yes, We will update your comment in version 3.0

→ **STATUS: *DONE*** *Password will be applied strong password mode, required to include uppercase letters, numbers and special characters*

- 'STAFF' account can create a user account, maybe restrict this function to only 'Super Admin'.
- *Our solution: Yes. We designed the system in a way that access rights to functions are managed through roles. A role is defined dynamically, with any name: Staff or Technical Team,... or similar, then you will be able to grant any function to this role. We/you can revoke any granted rights using the role management function. So for this issue, we just revoke the rights with a simple operation in the Role Management function available on the menu. During the Testing phase, we created them randomly. Before going live (after version 3.0 or 4.0,...), we will redefine the roles and their respective access rights to best suit the actual organizational model of MOF and UNDP. Of course, you can absolutely change it now.*

→ **STATUS: *DONE***

2. Role Management:

- Should have a clear guideline of who can create/read/update/delete which functions
- We would like to review the manuals to be developed by APIS.

Our solution: Thank you. We will continue to update the user manual with more complete content for your team.

➔ *STATUS: Our team (APIS) will update the new version of the user manual for you.*

3. Under the menu: Get Tax Data by importing from Excel and by API:

- When login with other account types (except Super Admin), these functions do not work. I assume they are intended to be handled by the 'Super Admin' ? (please correct me if I'm wrong).

Our solution: We are designing the system based on role-based access. It is possible that the user you are using (or have just created) has been assigned a role that is not allowed to perform this function. This can be done simply by changing the role or creating a new role with the permissions you want (or your rules). We may need to update the user documentation to better serve your team.

➔ *STATUS: DONE*

- If that's the case, these should probably be hidden for other accounts. The same should apply to any other functions that are not designated for certain account types.

Our solution: We will consider hiding / or notifying access to the function if an account is not granted access. We will update this comment in version 3.0

➔ *STATUS: DONE*

4. Under the menu: Get Tax Data by Import from Excel -> Data Requirements to estimate TE -> Profit Tax

- There's no option to edit/delete the specific line, as well as batch deletion feature for handling large amounts of data
- Maybe the 'TIN' information needs to be updated.

Our solution: Sorry, we have displayed too much information on one screen, so it is hidden on the right side. You can drag the scroll bar to the right to see it. Maybe in the future (in the next versions, based on the actual conditions of the users - or

your specific requests, we will hide some data columns to make it more convenient). We will update this issue in the next versions - if your team requests.

→ **STATUS: DONE**

Profit Tax

Dashboard - Profit Tax

Browse Data

(Show 1 - 10 data from total 100 records)

Search

Filter

Export

TIN %	ZONE 1 %	ZONE 2 %	ZONE 3 %	PROVINCE %	DISTRICT %	SECTOR %	TE NO %	REVENUE %	EXPENSE %	PROFIT %	BENCHMARK PT RATE %	BENCHMARK PT 2 N %	PT %	CUSTOMS 10 NO %	CUSTOMS 10 %	NET TE NO %	IMPORT NET TE %	TOTAL %	ACTION
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.24	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.24	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.2	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.2	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.2	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.24	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.24	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.2	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.2	/	0	55	0	34	0	0	  
0	0	1		Champosak	Passe City	Agriculture	0	/	/	/	0.2	/	0	55	0	34	0	0	  

Items per page: 10

Previous

1 2 3 4 5 6 7 8 9 10

Next

5. User activity tracking: edits, deletions, submissions, etc.

Our solution: Currently, because there is no specific request for this function, we have only temporarily developed the access log tracking function. We will implement this issue in the next version when there is a more specific request (for example, monitor / log what types of data are edited, what types of data are deleted... because I am sure that you will not want to store everything, you only need to store operations related to sensitive data. We will need more information from you - if any, to complete the function in the next version (version 3.0).

→ **STATUS: DONE – simple update** (Function on Menu System - > Operation Logs)

We will continue update when you have more comments in the future

6. Admin tracking and monitoring:

- Error log viewer (database error, etc.)

- Users logs for monitoring users transactions in the system, for security reasons and backup if any changes are made without intention. This should be monitored by the Admin users.
- Export logs to CSV, PDF, etc.

Our solution: *Yes, We will update your comment in version 3.0*

Of course we also need you to specify what types of transactions you want to track and what data types they will apply to. I assume you don't want to track everything (because there will be a lot of data being processed - for example when importing data via the API, the number of records processed will be huge. Storing all transactions for all data types will require a lot of server resources. You may want to consider this. We will update them after we have more specific requests from you

→ **STATUS: DONE – simple update** (Function on Menu System - > Operation Logs)

We will continue update when you have more comments in the future

7. Backup and Restore:

- In case a user accidentally deletes data or for other purposes.

Our solution: *Yes, We will update your comment in version 3.0 or 4.0 (as included in the development plan). We will create background backups at times when the system is most idle (e.g. at midnight, 2-3am). Of course, you can change these values whenever you want by simply using the menu function. We will also add the function of manual ad-hoc backups to the menu - however, we recommend you to choose the right time to execute this function, because it will slow down the system significantly, especially when the system has a lot of data. Data recovery also requires a lot of server resources. However, we will still put those functions on the menu. The implementation will be decided by you (the user). You can rest assured about that.*

→ **STATUS: DONE** Currently we have set up a prototype daily backup task at 2am. These backups will be overwritten after 30 days (for server resource reasons, in the future you can change to store them for months, years... when you have enough resources for this requirement). The directory where these backups are stored is: /var/backups/mysql

We also provide a full system backup function using the Menu function. Users can perform a backup using the UI function and can copy the data to their personal computer. Of course, you have the right to limit who can perform this function. This is done through the mechanism of using the function rights for each role (like the system currently has). Of course, we can change it to not allow copying backups to personal computers - if you want.

Function on Menu System - > Backup / Restore

- Prepared for sustainability of data available for the long term when data bandwidth might increased in the system and causes a slow searching/responding.

Our solution: System optimization is always a difficult problem for any information system and always requires appropriate resources (human resources and hardware resources). No system can guarantee that it will always run well and quickly automatically when data increases without optimization activities from experienced administrators (especially for systems that require development in the direction of using open source code and free databases (MySQL, Postgre, ...)). TaxETS is no exception. Therefore, in this matter, we will try to arrange resources to support you during the warranty period. After that time, you should renew the maintenance contract or hire another company to perform this task or send your technical staff (if any) to attend in-depth training courses on information systems to be able to master and perform monthly / annual system optimization work yourself.